

Doctoral Dissertation Defense

学位（博士）論文公聴会

# A Study on Badminton Video Analysis Based on Deep Learning Models

（ 深層学習モデルに基づくバドミントン映像解析の研究 ）

Graduate School of Systems Design, Department of Electronics and Information System Engineering

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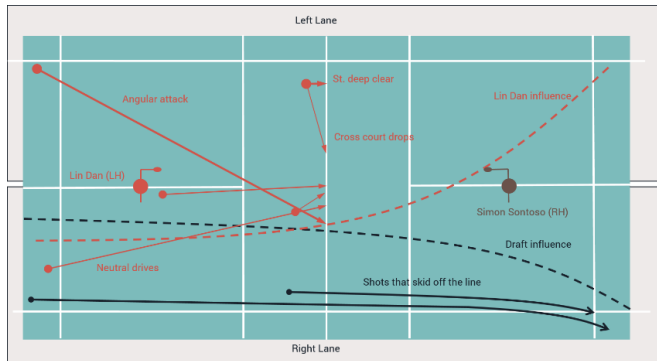
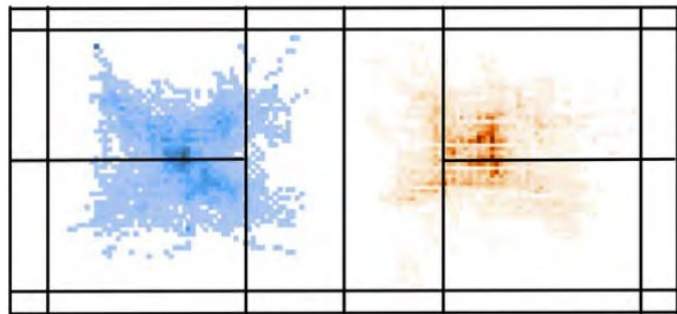
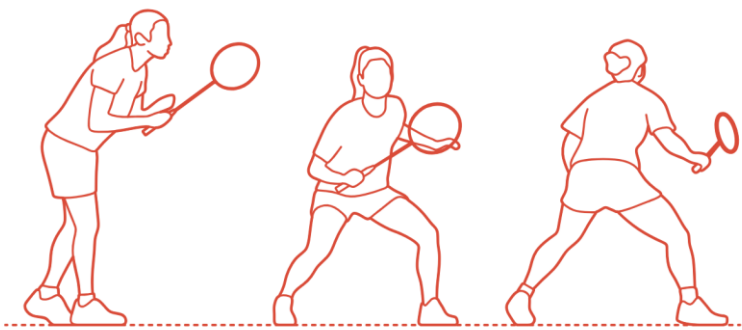
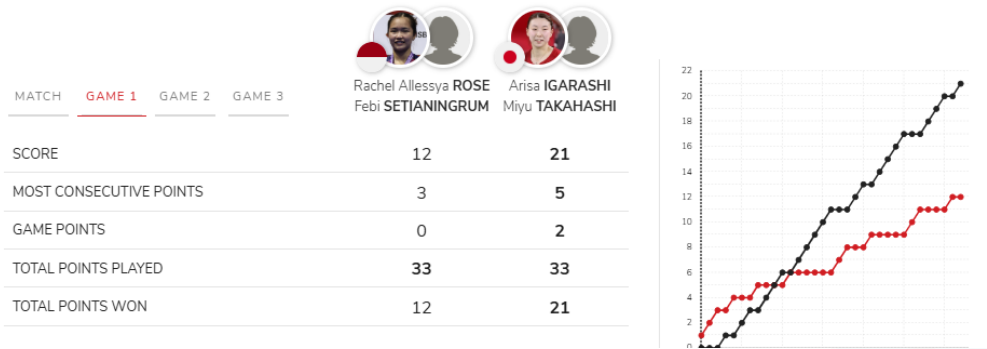
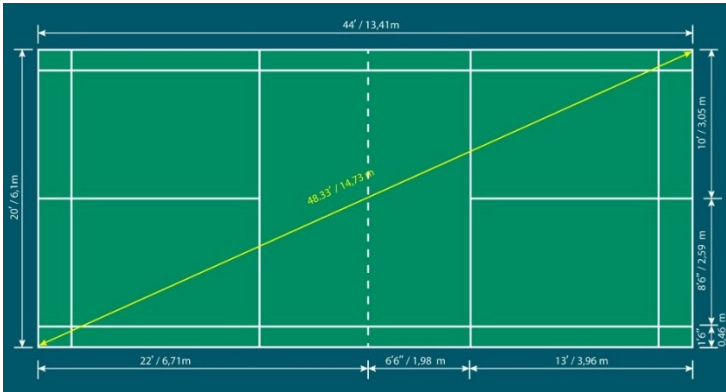
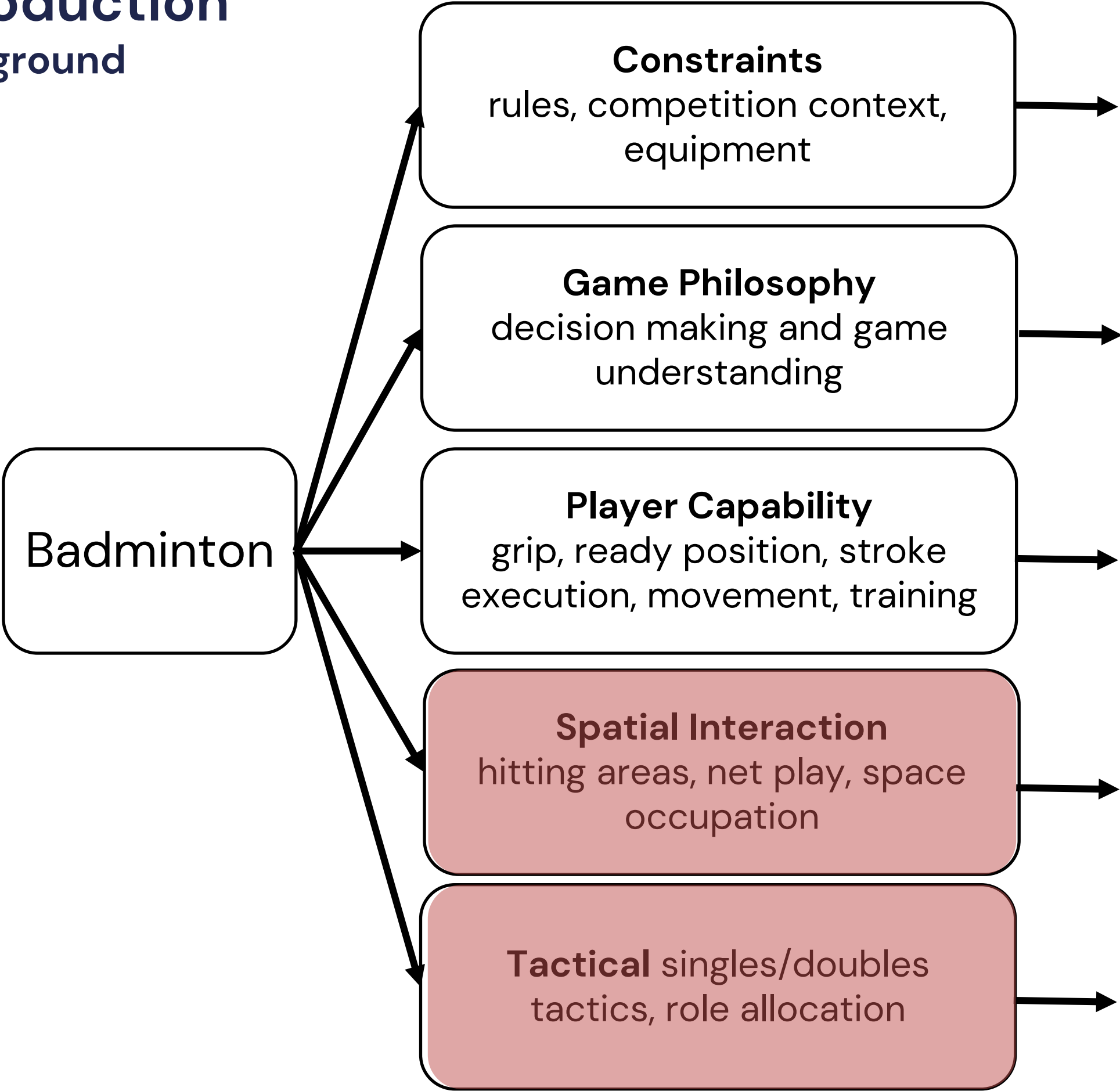
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Supervisor: Prof. Norio Tagawa

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# Outlines

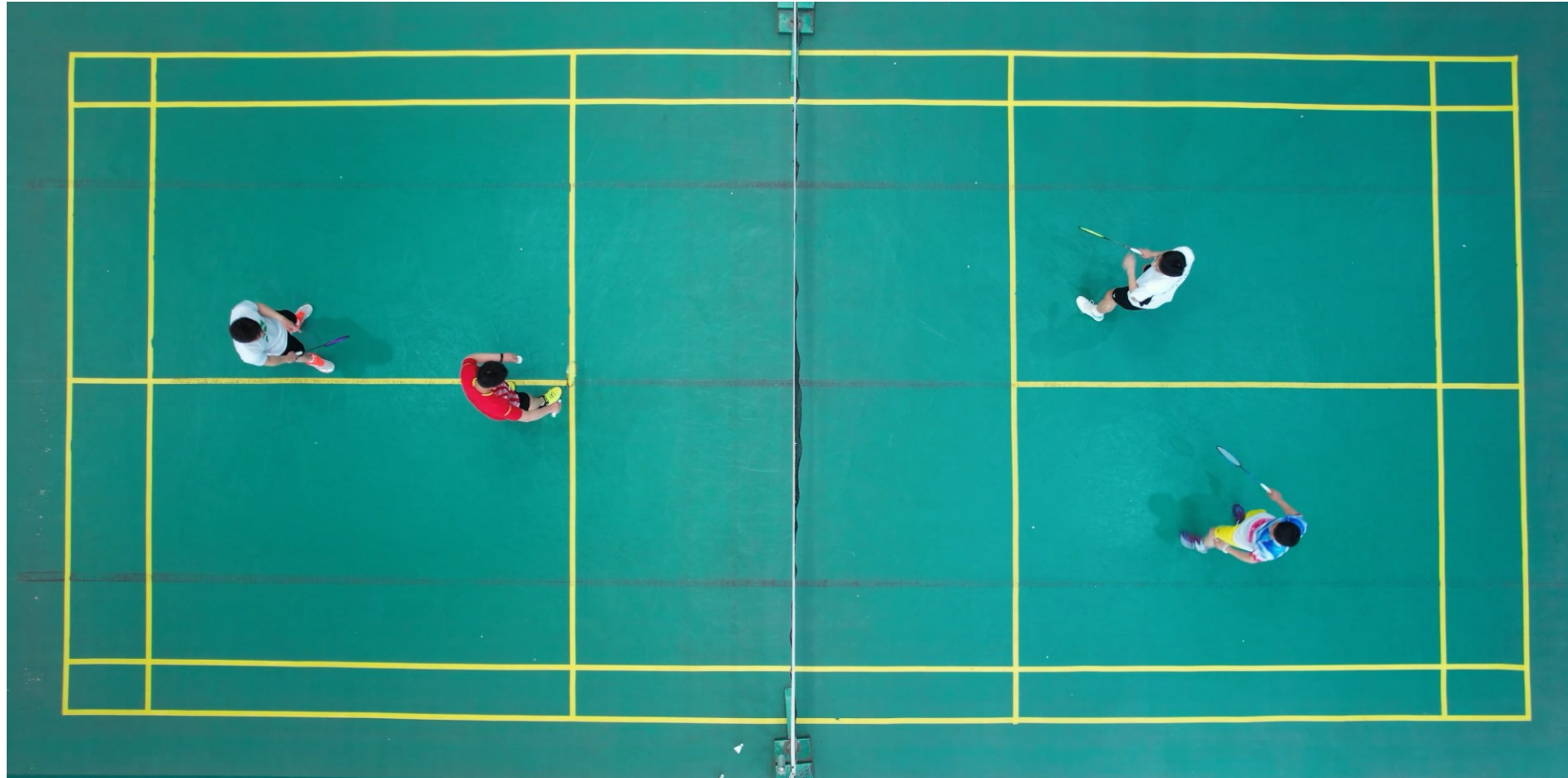
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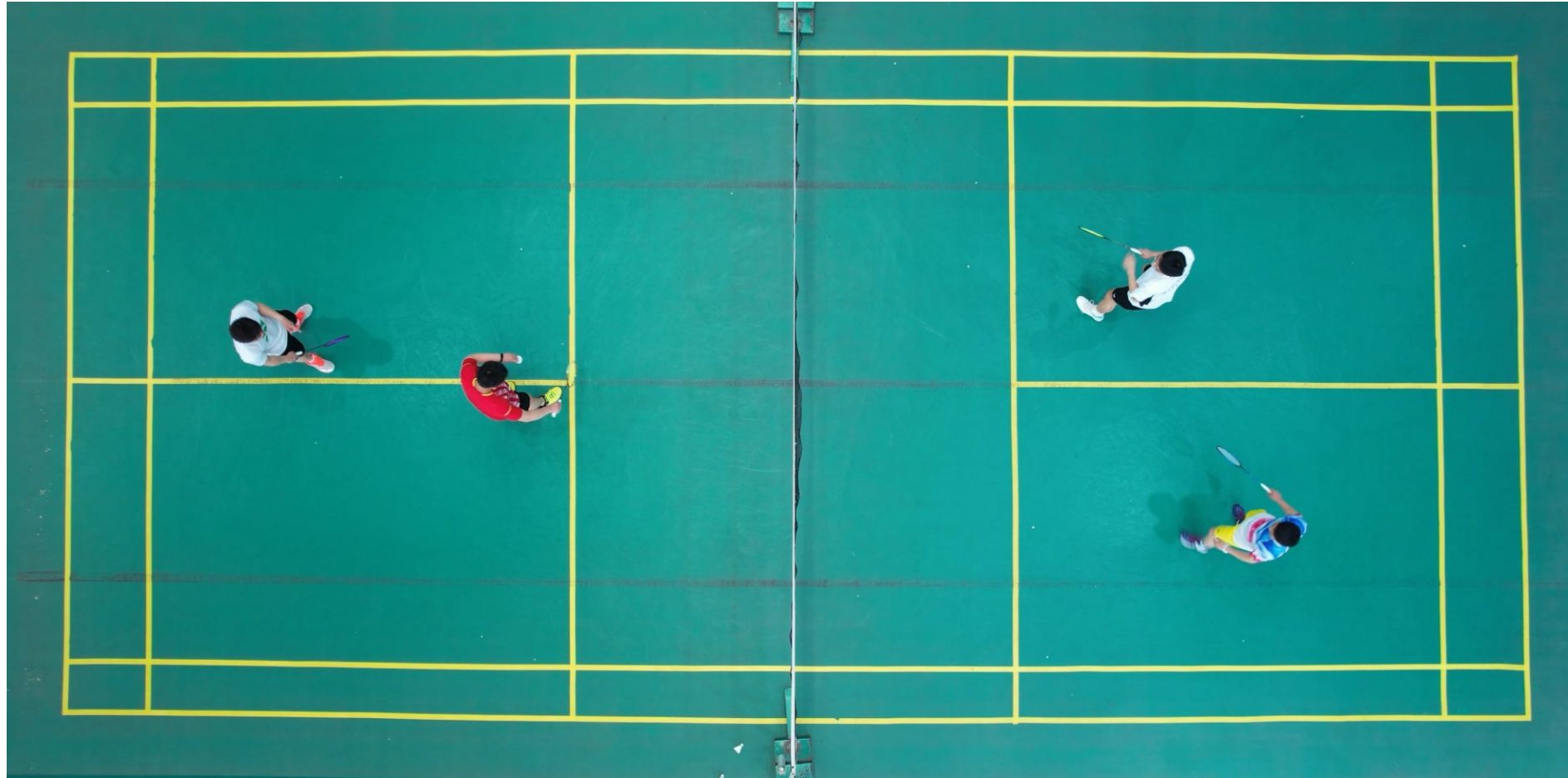
# Chapter 1

## Introduction

### Background (Cont.)



- Badminton is a fast racket sport with rapid player movement, high-speed shuttlecock motion, and frequent interactions among players, rackets, and the shuttlecock.
- **Tactical understanding**, especially in doubles matches, depends on spatial positioning, court coverage, and player coordination.



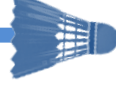
- Although computer vision has advanced sports video analysis, badminton remains **less explored** than other sports.
- Challenges such as the shuttlecock's small size, high speed, racket rotation, occlusion, and complex interactions make analysis difficult, and many existing methods focus only on **isolated detections** rather than overall spatial dynamics.

Most existing badminton analysis focuses on **individual detections** and does not explicitly model spatial control on the court, which is crucial for tactical understanding in doubles play.

The objectives of this study are:

- Develop a pipeline for detecting and tracking **players, shuttlecock, and rackets**.
- Incorporate **oriented racket detection** with spatially meaningful representations.
- Introduce a **dominant area** visualization that represents court control by integrating player movement, shuttlecock position, and racket information.

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Rahmad et al. (2019) introduced an automated badminton player detection method using **Faster R-CNN** on broadcast match videos. Players were detected from extracted image frames using bounding boxes

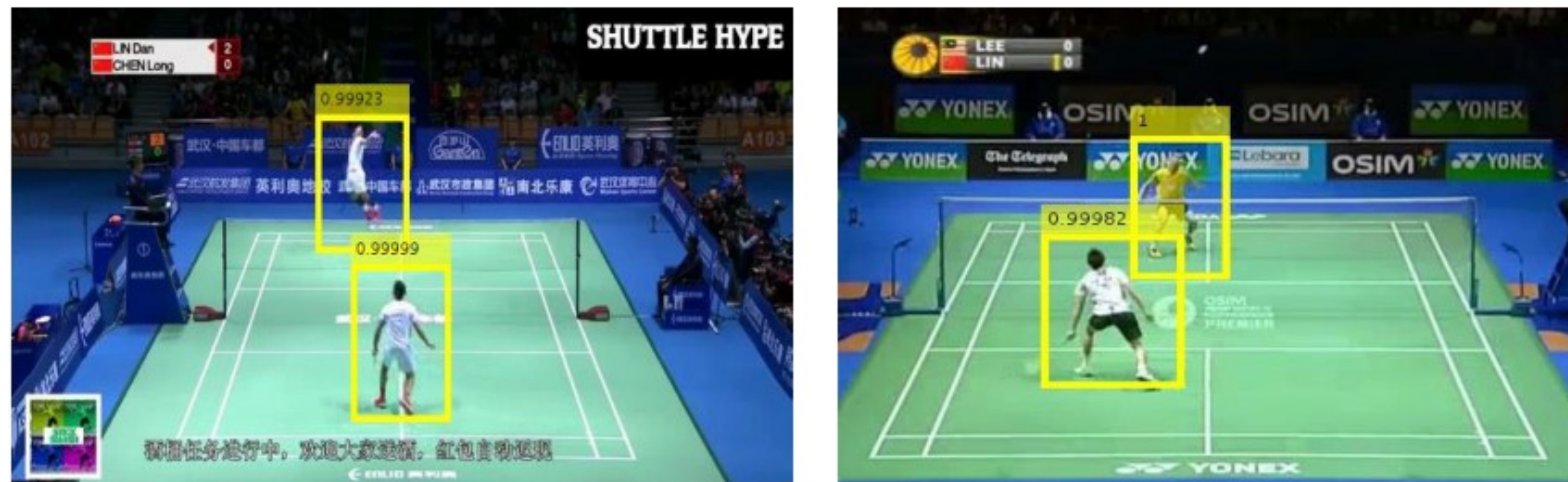


Figure 2.1. Example results of badminton player detection from broadcast match videos using Faster R-CNN

## Chapter 2

# Literature Review

### TrackNetV2

TrackNetV2 was proposed to detect the shuttlecock from badminton broadcast match videos (Sun et al., 2020). It consists of a **U-Net based encoder decoder** backbone and the layer is reduced with temporal input.

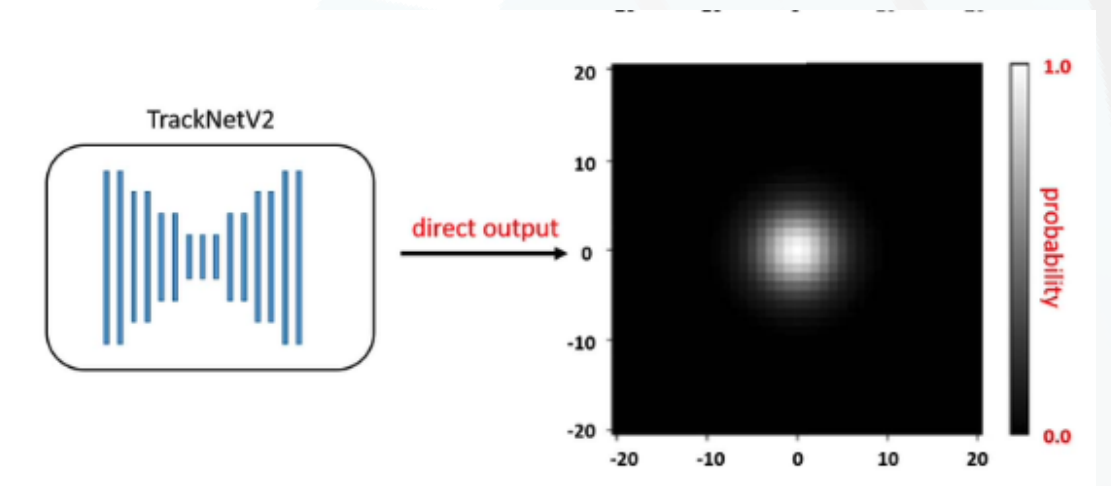
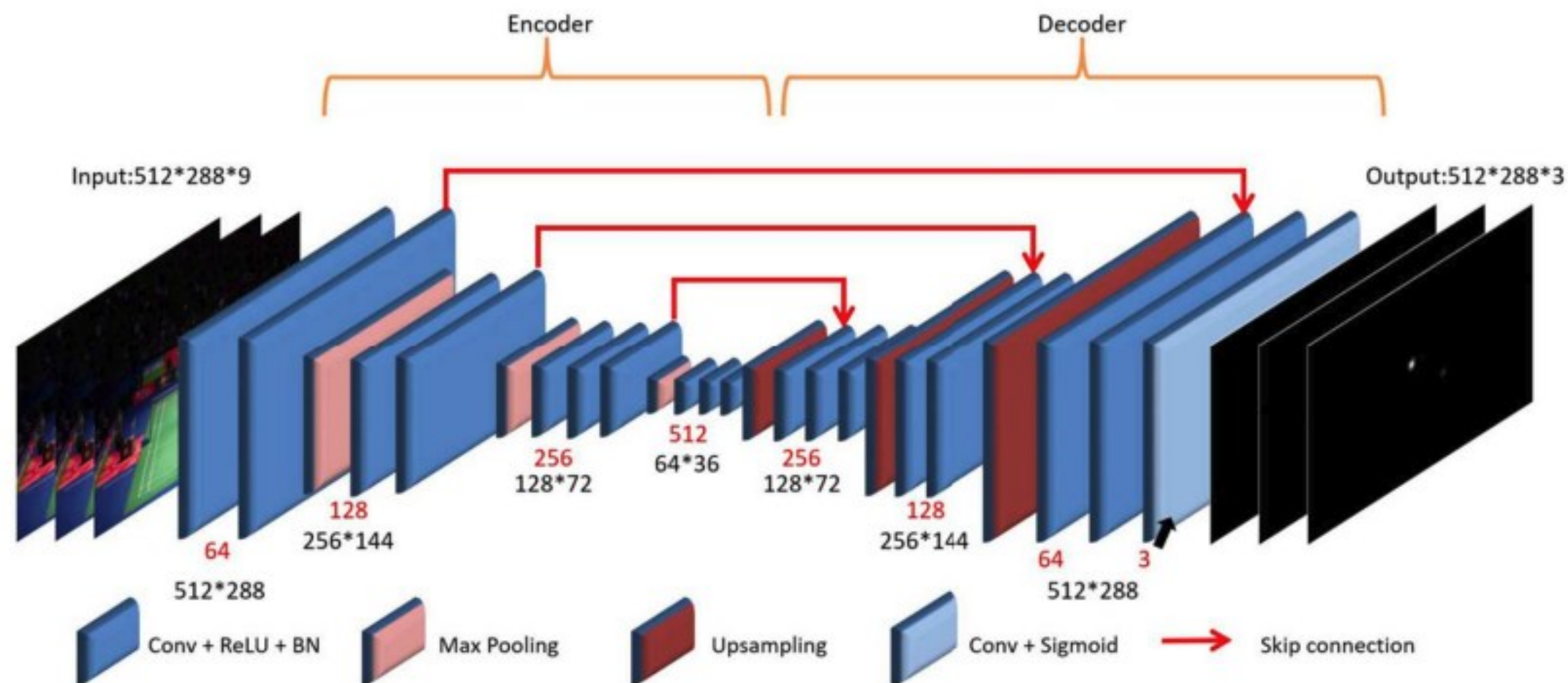


Figure 2.3. TrackNetV2 neural network architecture for detection of shuttlecocks

Figure 2.4. TrackNetV2 directly generates a real-valued array to have the heatmap.

## Chapter 2

# Literature Review

### VIRD

VIRD is Immersive VR based badminton match analysis system that integrates 3D reconstruction from video (Lin et al., 2023). It enables a top-down, interactive analysis workflow that supports tactical understanding. Anyway, no one research about racket in badminton when making this dissertation .



freely in the virtual court and watch the game from different angles. Further, all of them were excited seeing the moving bird and the 3D reconstructed game. Throughout the analysis, experts felt engaged and interactive, *"I like how I can move my body around and face the shot. I find that more beneficial than just looking at a TV screen or computer"* (C1). Experts also expressed their tendency to view a match in 3D and refer to the video only when the 3D view did not make sense. A coach also suggested adding rackets to improve the 3D view. Using VIRD, experts obtained deeper insights on the spatial aspects in their analysis. P1 observed most of his winners were hit downwards from his backhand side from analyzing the shot locations and trajectories.

Figure 2.5. Display of VIRD with the match animation from badminton match

- **Player control area** analysis was originally developed in football to represent pitch regions that a player **can reach** before any other player (Caetano et al., 2021).
- This concept provides a foundation for extending player area analysis to other sports, including badminton. But, we still need other information like **shuttlecock and racket** to build area analysis for badminton

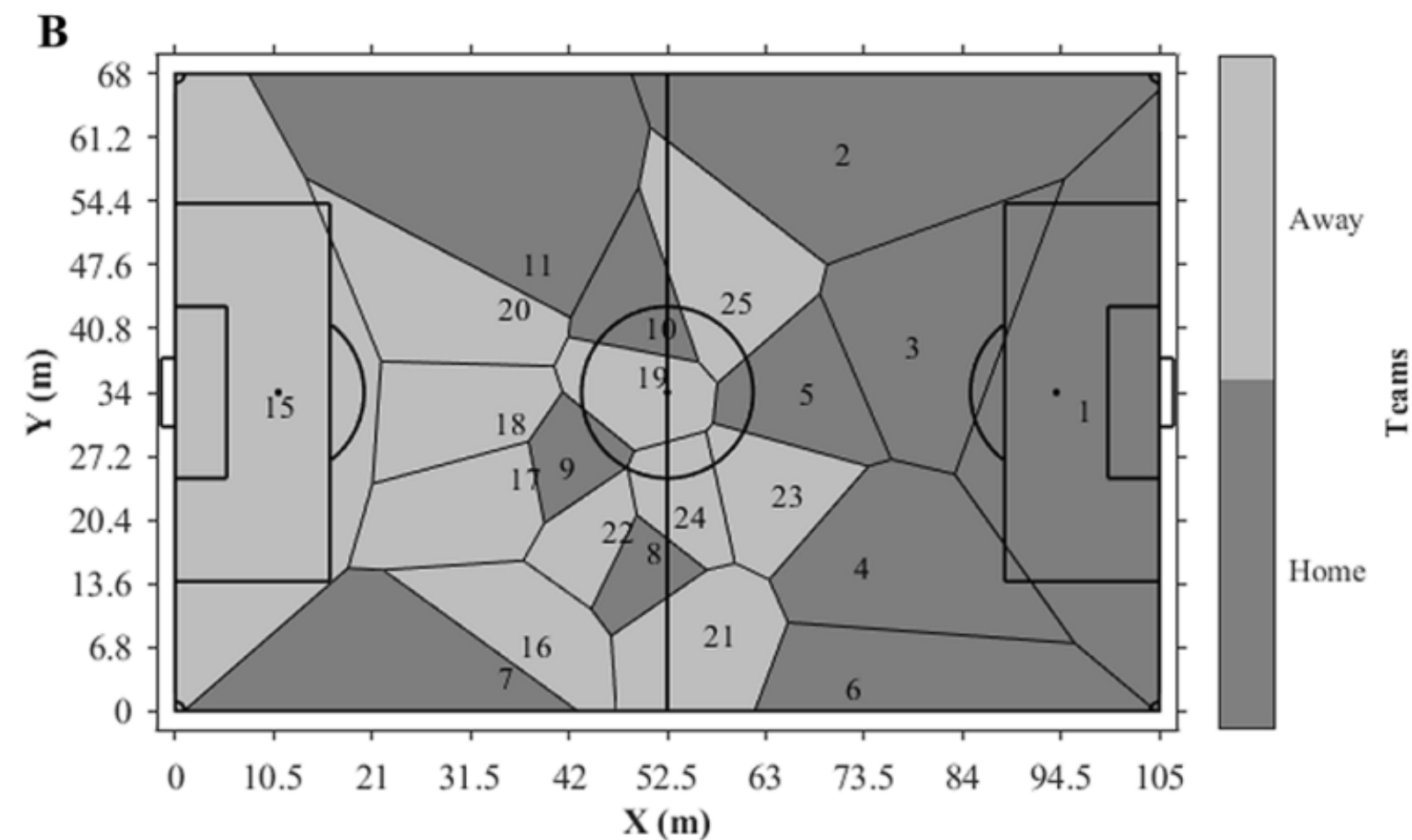


Figure 2.5 Example of player regions in football computed using Voronoi regions.

# Outlines

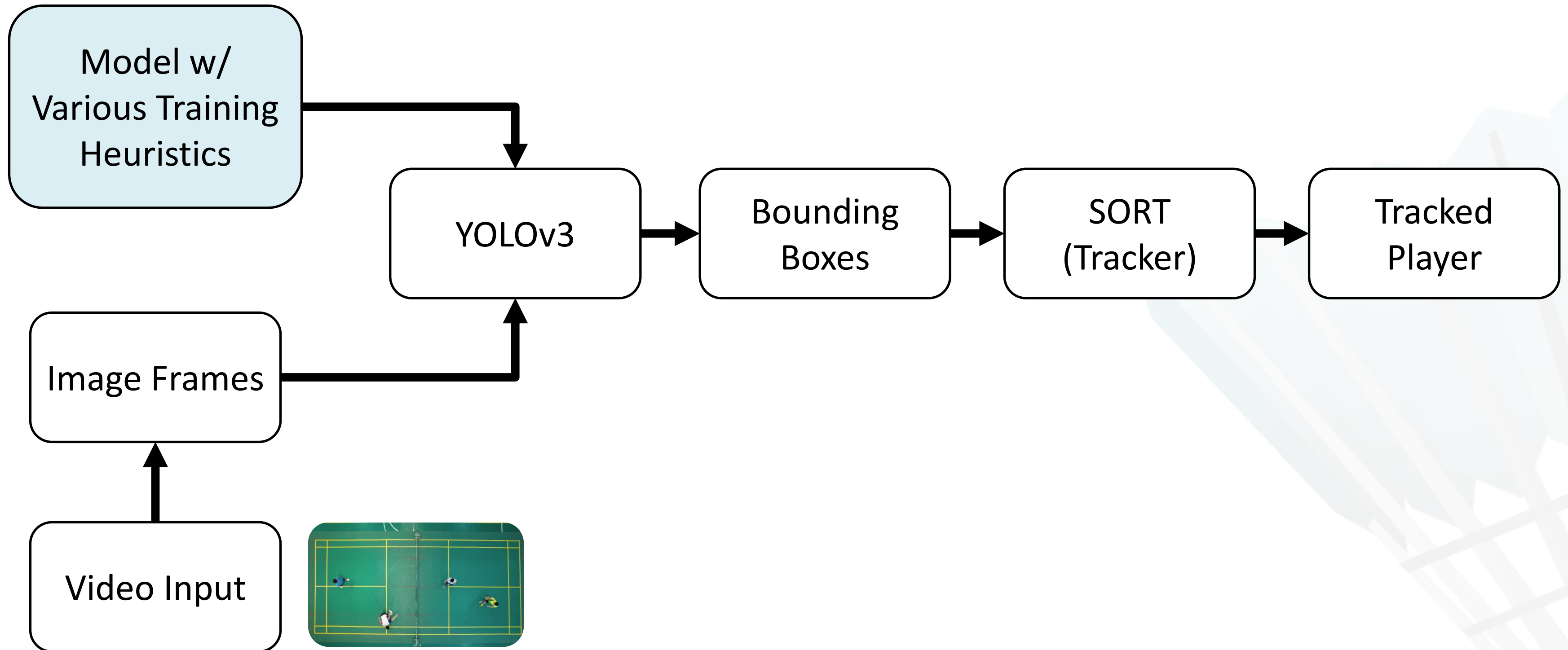
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- Accurate player detection in badminton videos remains challenging due to **rapid player movements, frequent occlusions, and significant scale variations**.
- These factors often lead to **unstable training** behavior and **inconsistent detection** performance when using standard training settings.
- To address this problem, this study investigates **various training heuristics** applied during model training.
- The objective is to **improve training stability** and **detection robustness** without modifying the underlying network architecture.

# Chapter 3

## Player Detection

Method : Pipeline



Apply Normalization, Cropping, Contrast Adjustment, HSV Augmentation to make the model more robust to different lighting, sizes, and positions.

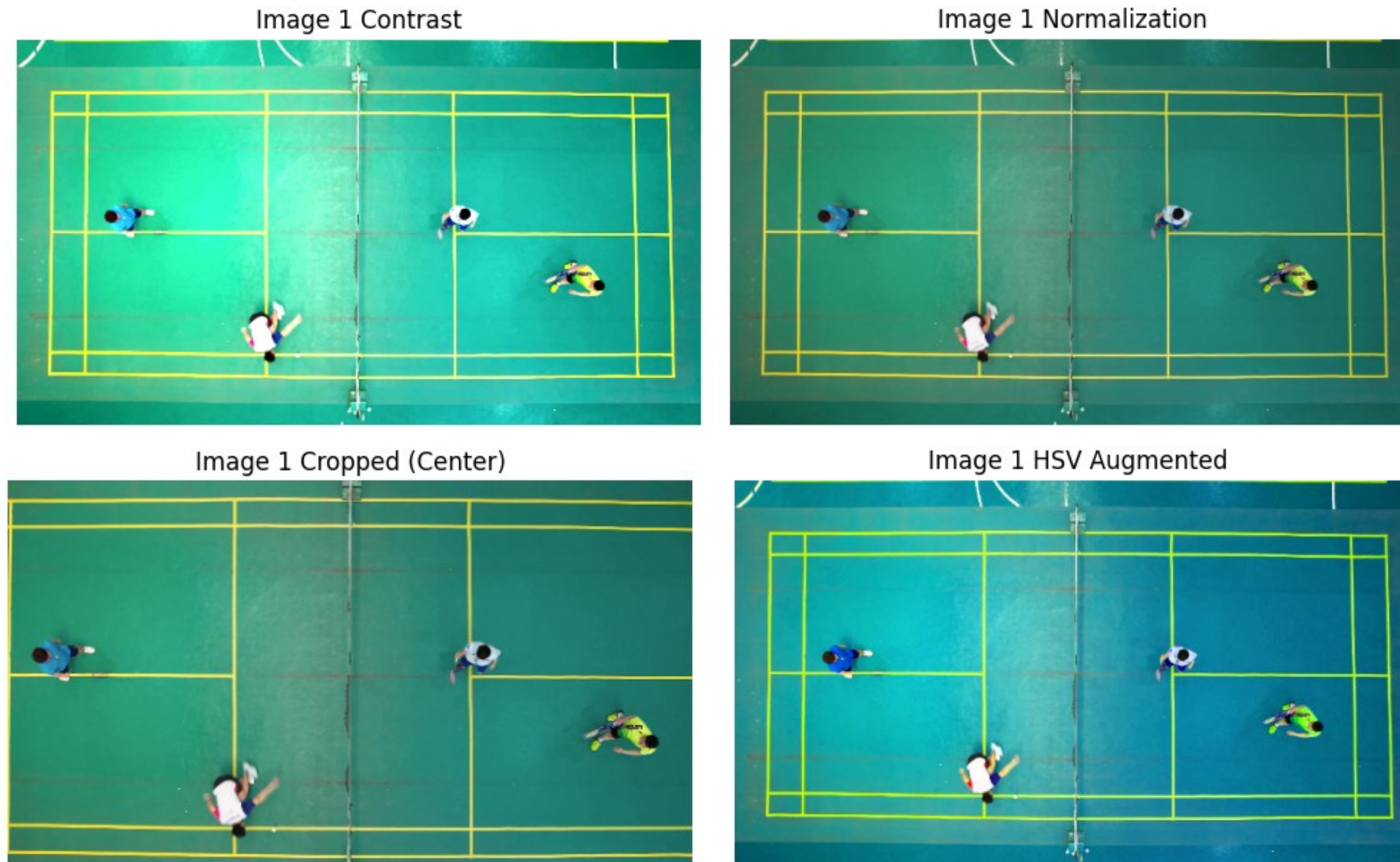


Figure 3.1. Data preprocessing and data augmentation used in player detection.

Cosine Learning Rate Scheduler, adjust the learning rate during training with cosine scheduler to make learning more **stable** and **accurate**.

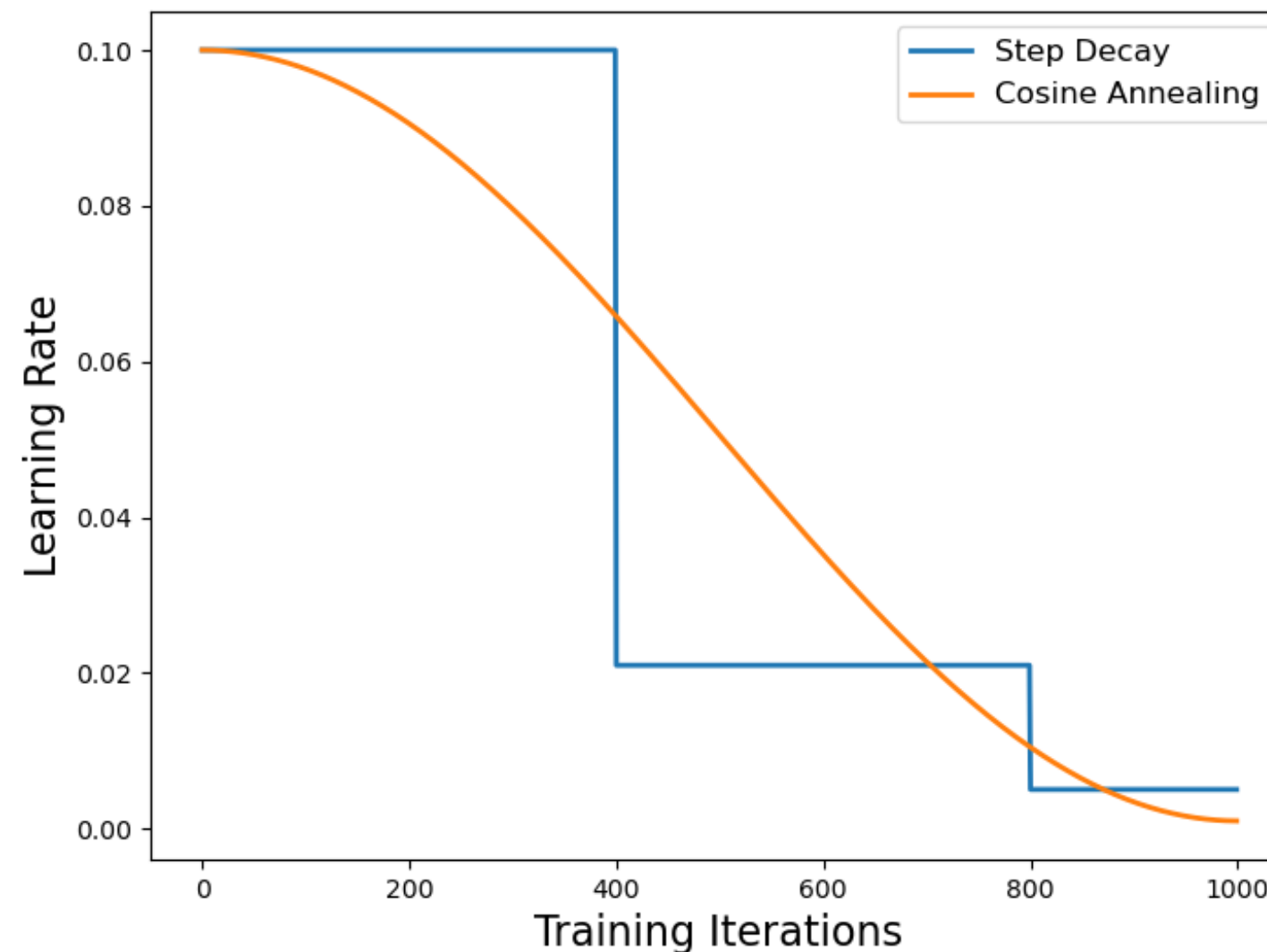


Figure 3.2. The difference between step decay and cosine learning rate.

Synchronized Batch Normalization (BN), share batch normalization statistics across GPUs to improve **consistency** and **speed up training**.

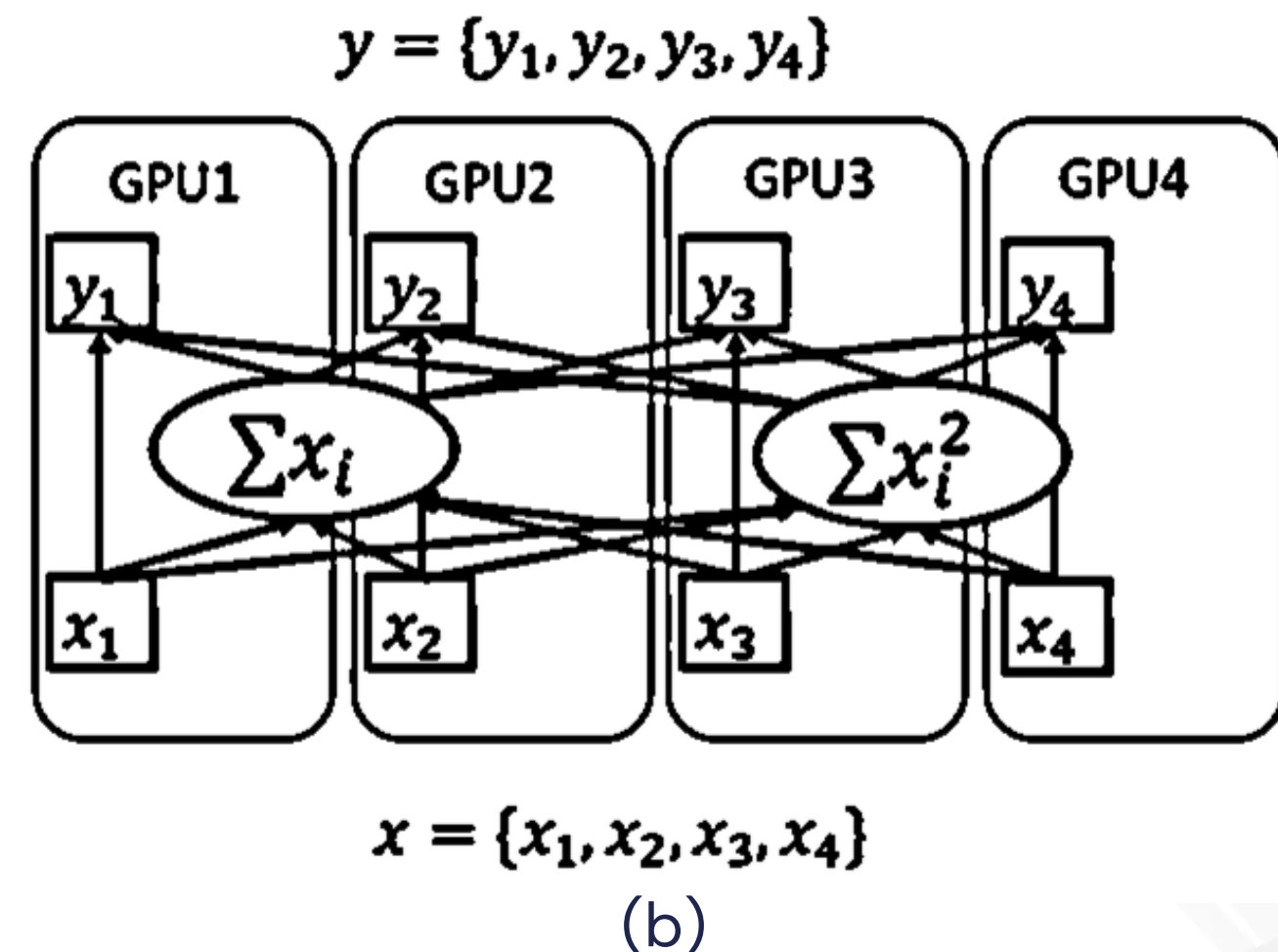
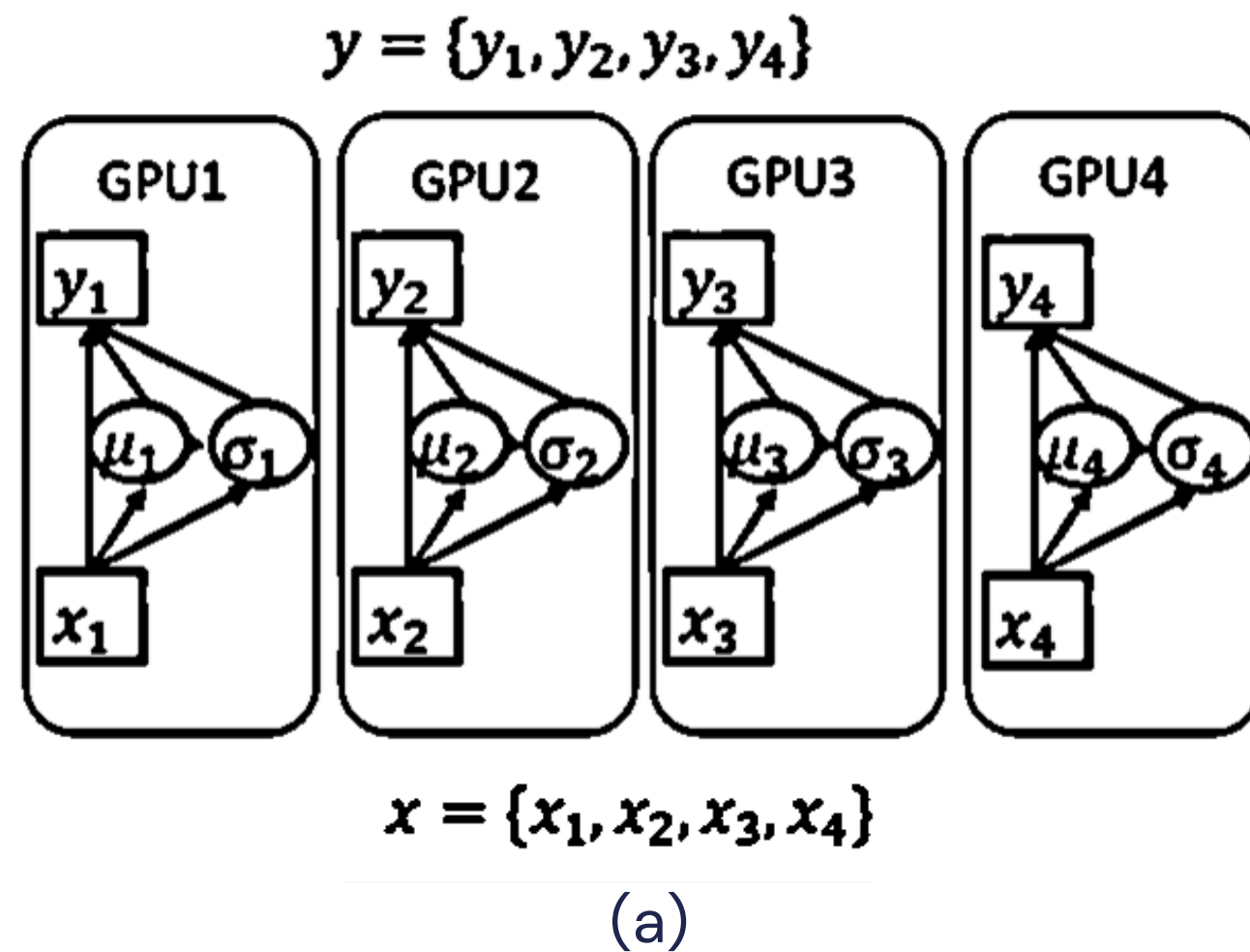


Figure 3.3. The difference between batch normalization (a) and synchronized batch normalization (b)

Combine two images into one by **blending their pixels**. This makes the training data more **diverse** and improves detection accuracy.

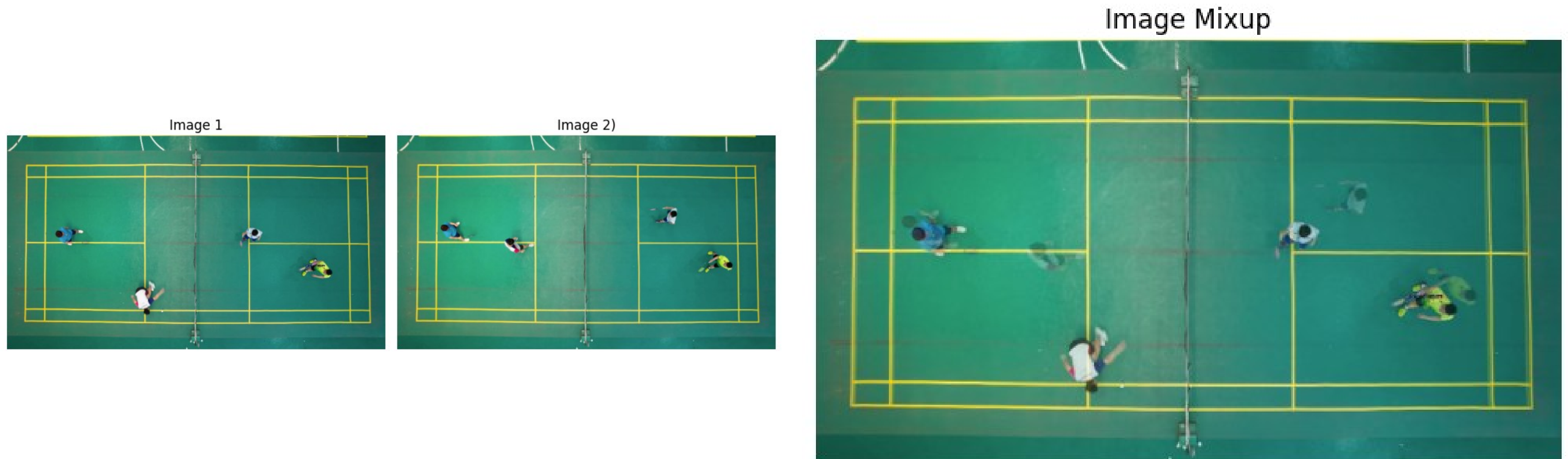


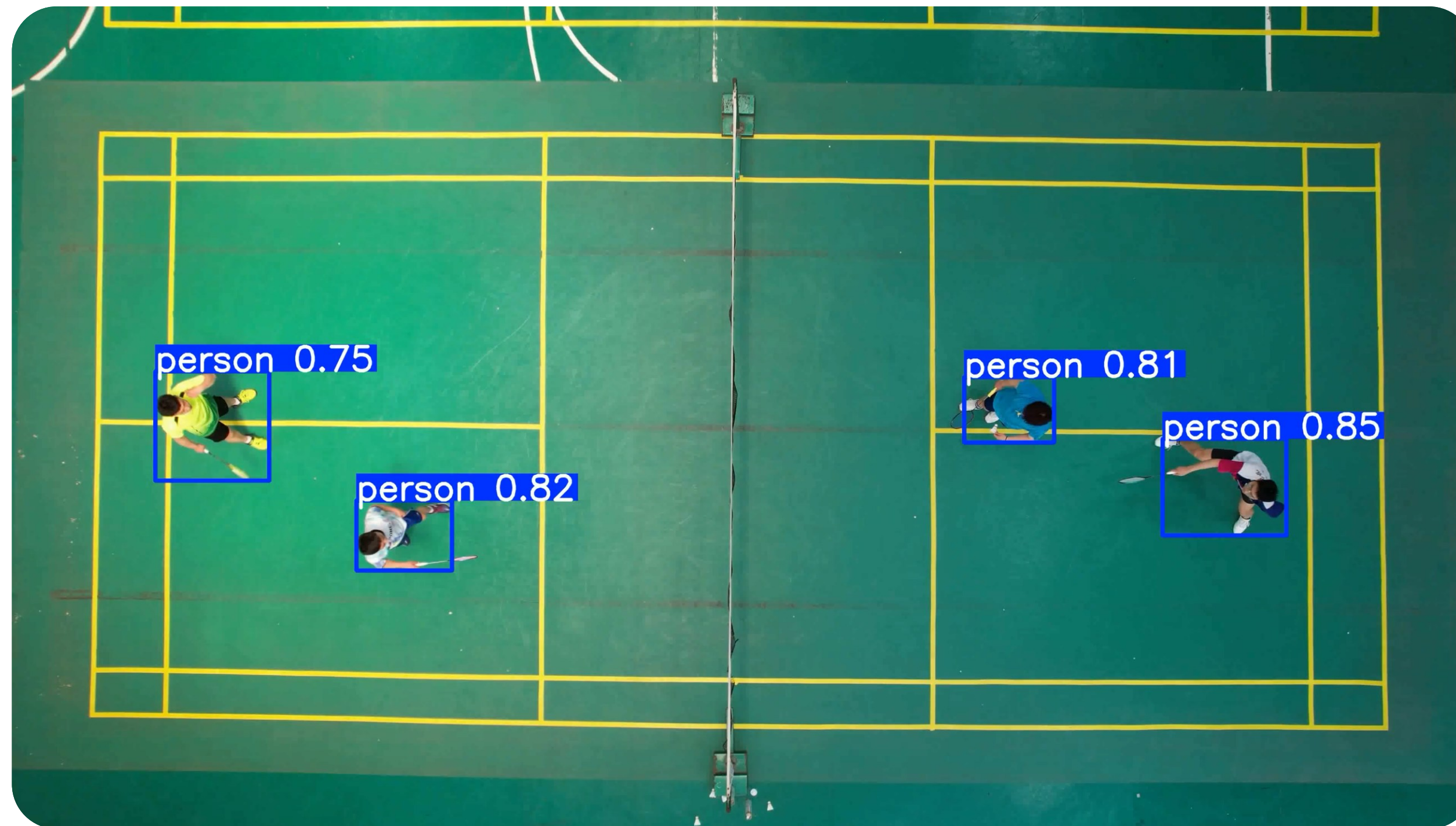
Figure 3.4. Example of Image Mixup that generated by linearly combining pixel values from two input images.

Performance comparison of player detection on the COCO dataset based on mAP@50–95.

| Various Heuristics    | mAP   | $\Delta$ | Cumu $\Delta$ |
|-----------------------|-------|----------|---------------|
| Baseline              | 33.05 | 0        | 0             |
| + Data Pre-processing | 33.58 | 0.53     | + 0.53        |
| + Cosine LR Schedule  | 34.52 | 0.94     | + 1.47        |
| + SyncBN              | 35.67 | 1.15     | + 2.62        |
| + Image Mixup         | 37.35 | 1.68     | + 4.3         |

Table 1. mAP comparison after various heuristics training.

## Baseline + Various Training Heuristic

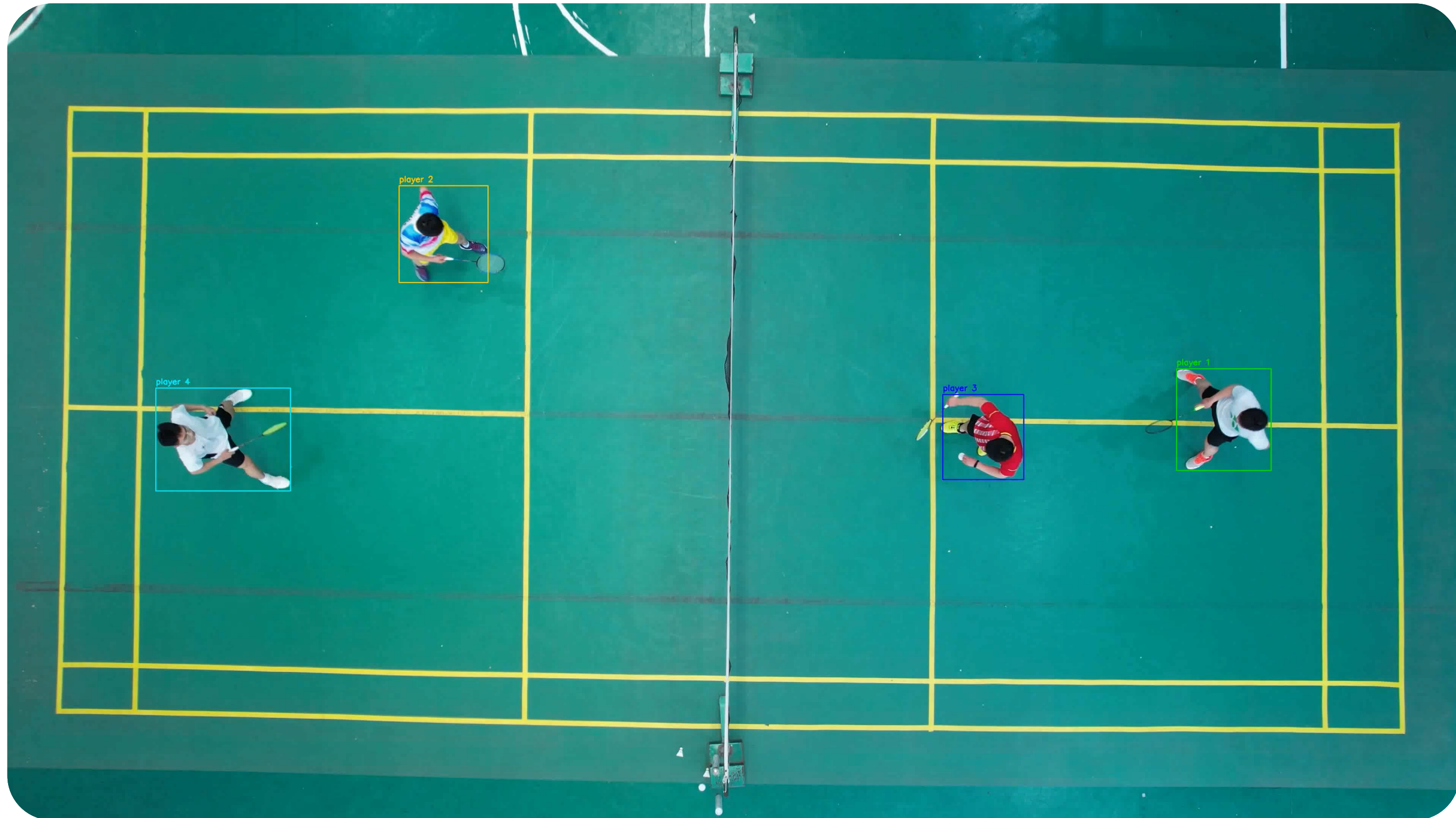


# Chapter 3

## Player Detection

### Result (Cont.)

## Baseline + Various Training Heuristic + SORT (Tracking)



Chapter 3

# Player Detection

Result : Player Heatmap Statistics

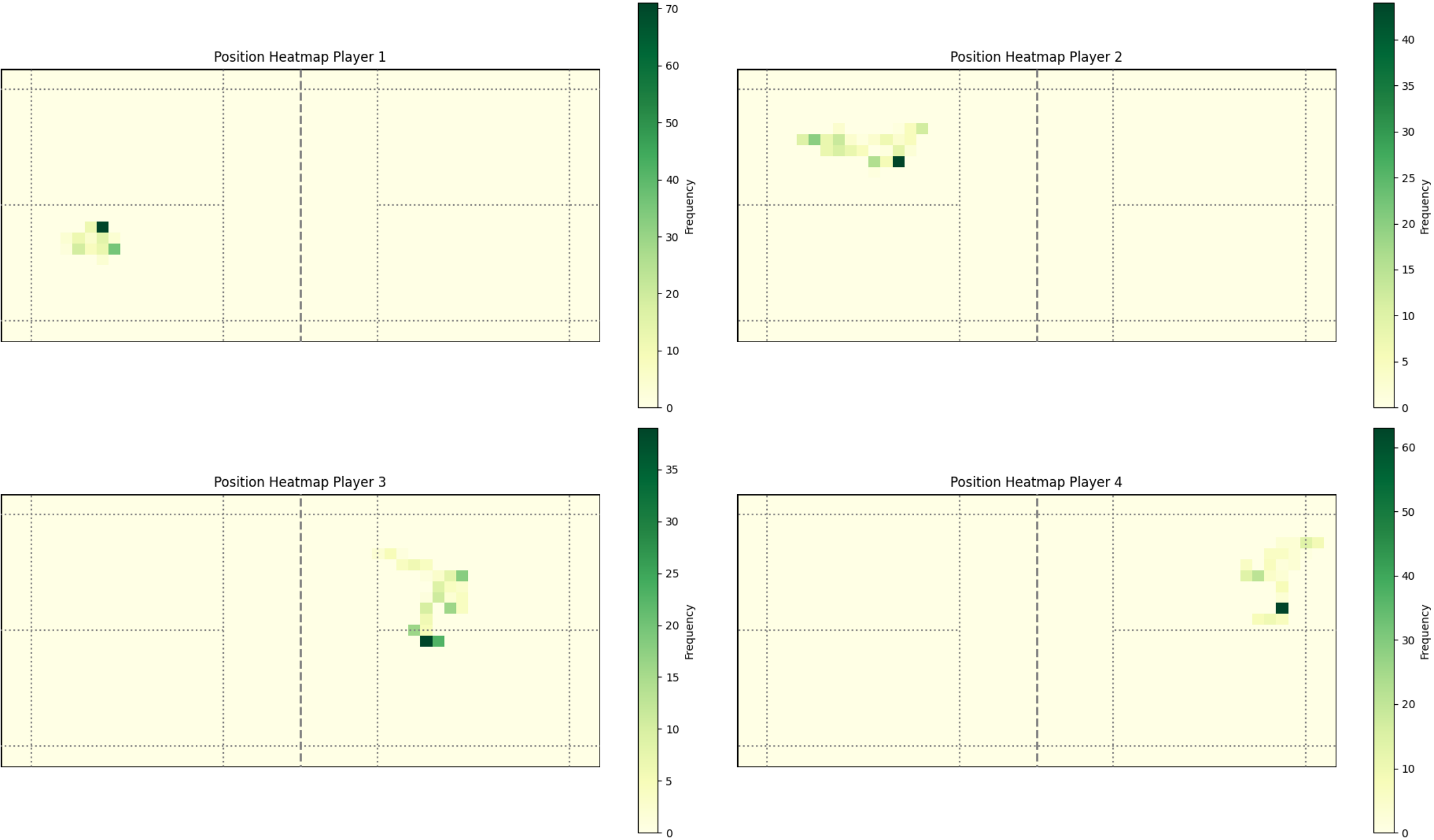


Figure 6.3. position heatmap for each player in badminton double matches.

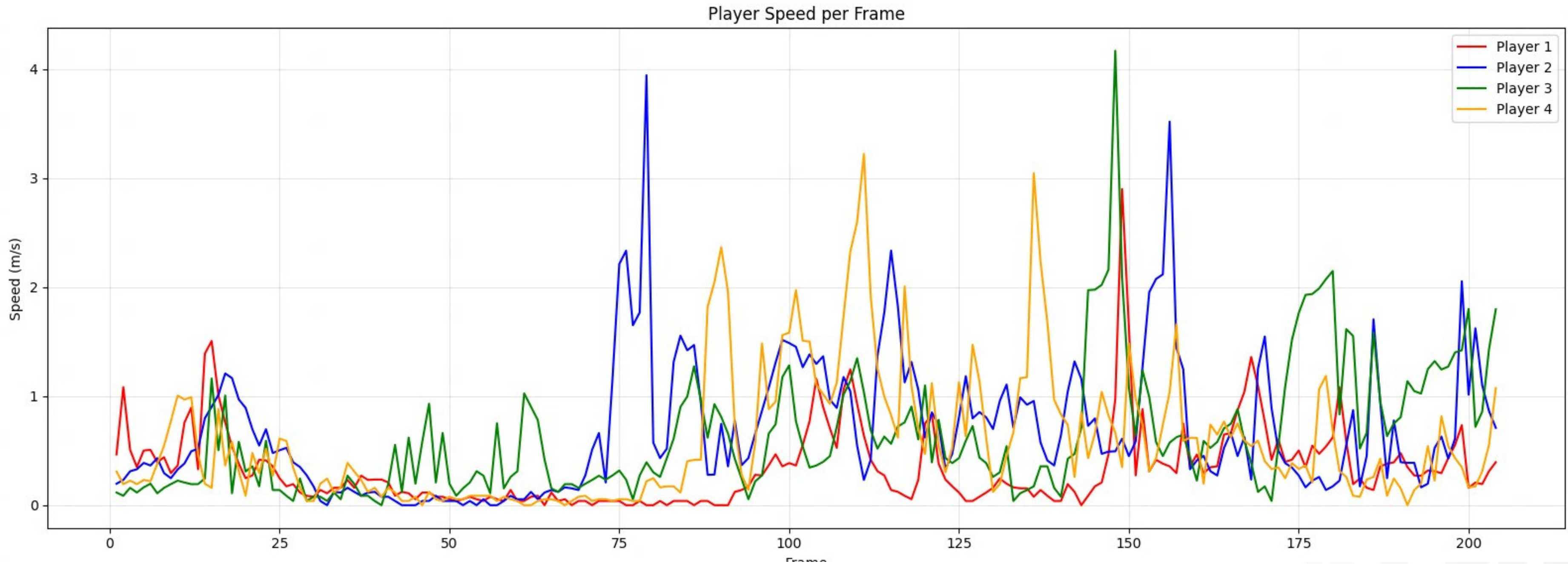
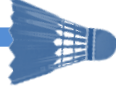
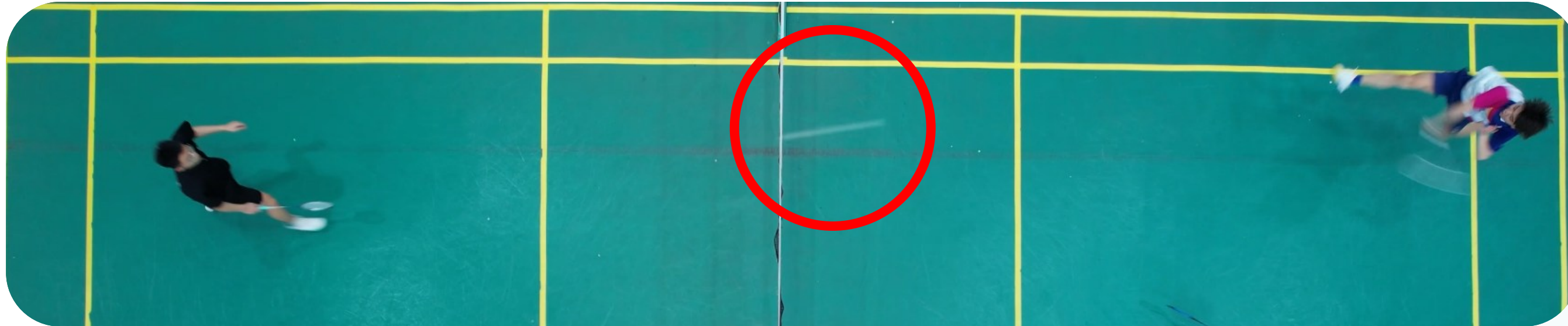


Figure 6.4. Speed statistics for each player in badminton double matches.

- Player detection performance was improved using heuristic training strategies on YOLOv3.
- Applied techniques include data preprocessing and augmentation, image MixUp, SyncBN, and cosine learning rate scheduling.
- Reliable player trajectories were obtained by integrating SORT tracking, enabling stable spatio-temporal analysis.
- mAP increased from 33.05 (baseline) to 37.35

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- Shuttlecock detection is a **challenging** task due to its **small** size, **fast** motion, and frequent **motion blur** in badminton videos, which often lead to **unstable predictions**. Even with a strong baseline (TrackNetV2), detection performance can degrade in fast rallies and complex motion patterns.
- To address this problem, this study builds upon **TrackNetV2** as a **baseline** by introducing **residual connections**, replacing ReLU with **Leaky ReLU**, and repositioning **batch normalization after residual addition**. These modifications aim to tackle **vanishing gradient problem** and improve training stability.

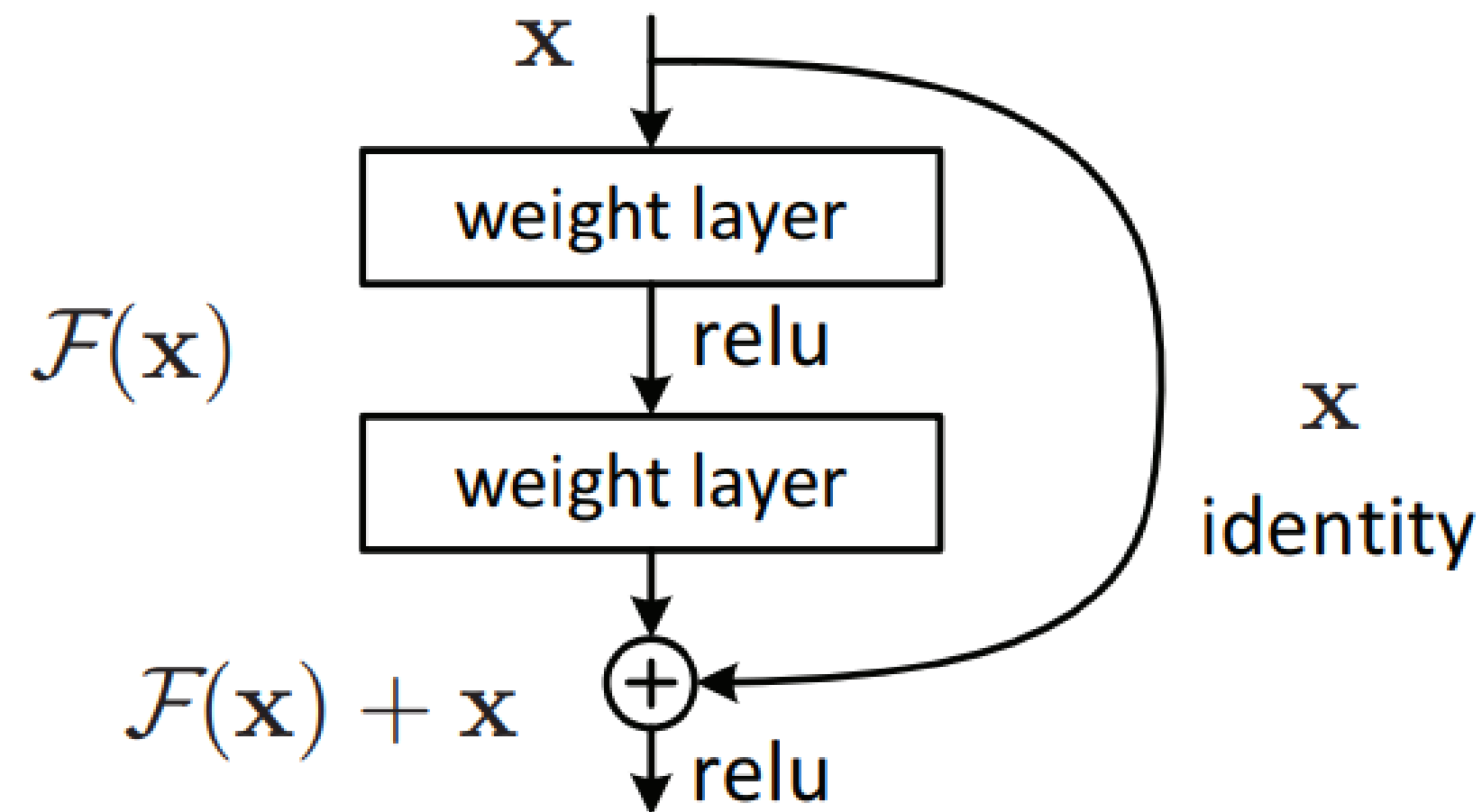


Figure 4.1. Residual block architecture illustrating the skip connection, where the input  $x$  is added to the residual mapping  $F(x)$

Proposed the extended version of baseline (TrackNetV2) by introducing **residual connections** (red arrow) in its blocks. Residual connection could mitigate the **gradient vanishing problem**, resulting in higher shuttlecock detection performance.

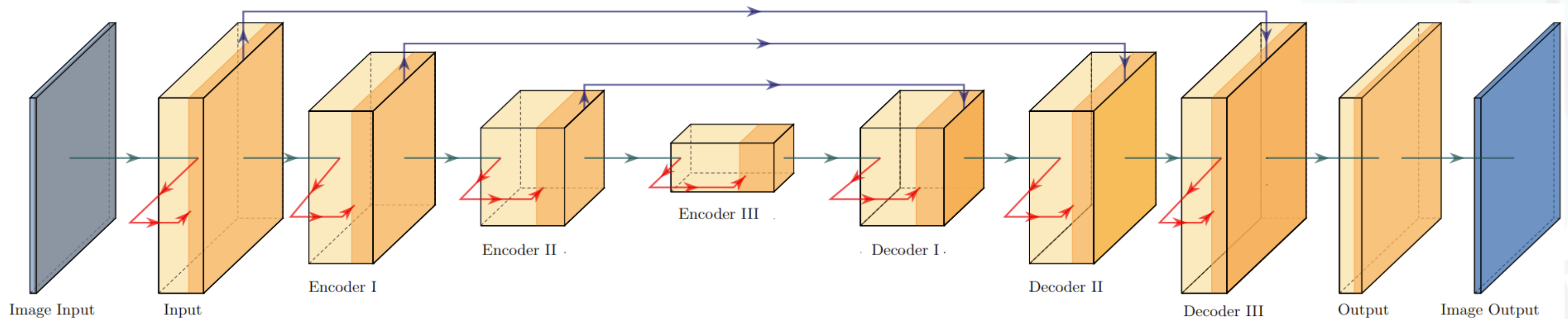
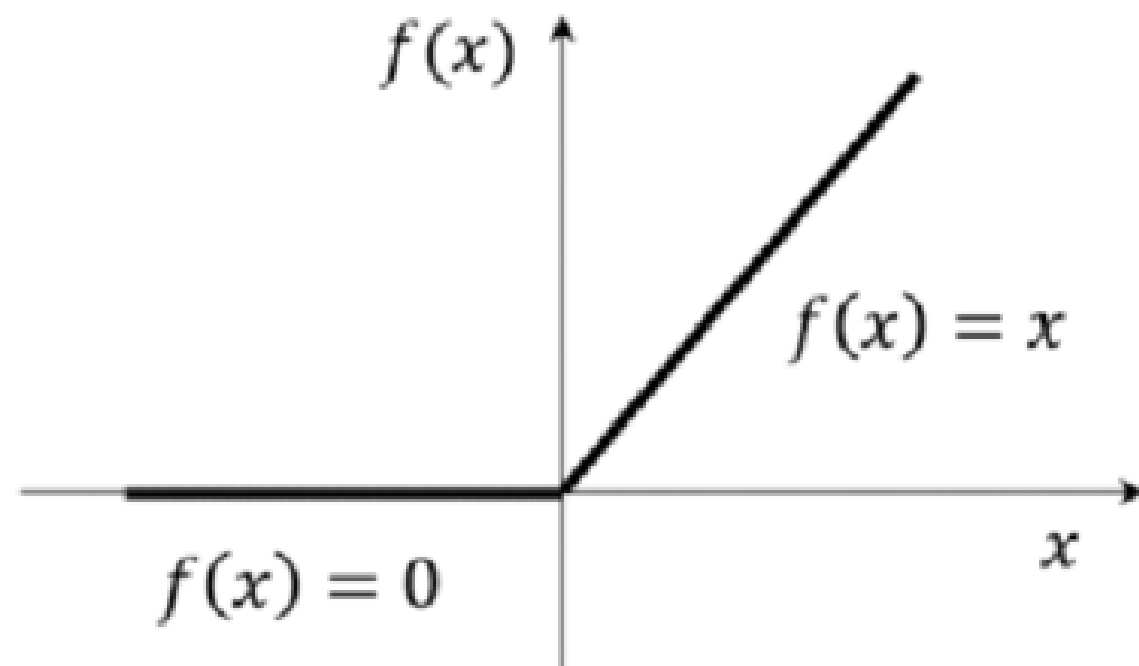
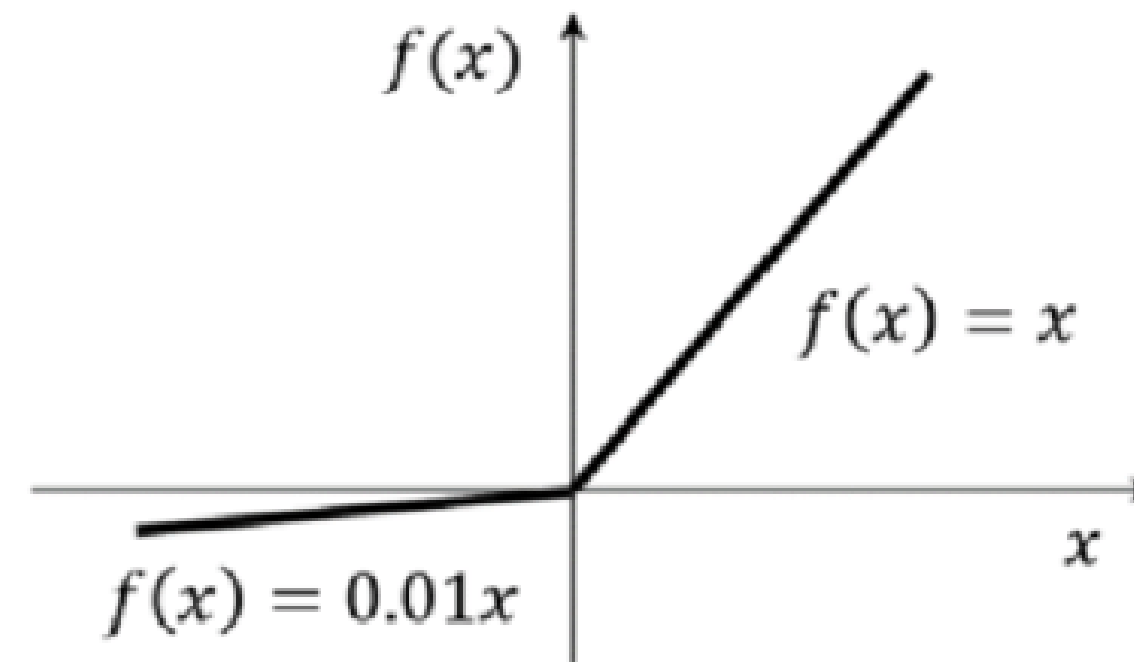


Figure 4.2. Overall network layer detail of shuttlecock detection system with the addition of residual function.

The shuttlecock is small, fast, and often appears faint or partially blurred. ReLU **suppresses negative activations**, which can remove weak but important features. This **increases** missed detections when the shuttlecock signal is subtle.



*ReLU activation function*



*LeakyReLU activation function*

Leaky Rectified Linear Unit is used to address the **dying ReLU problem** by preventing some neurons from turning inactive and stopping learning during training.

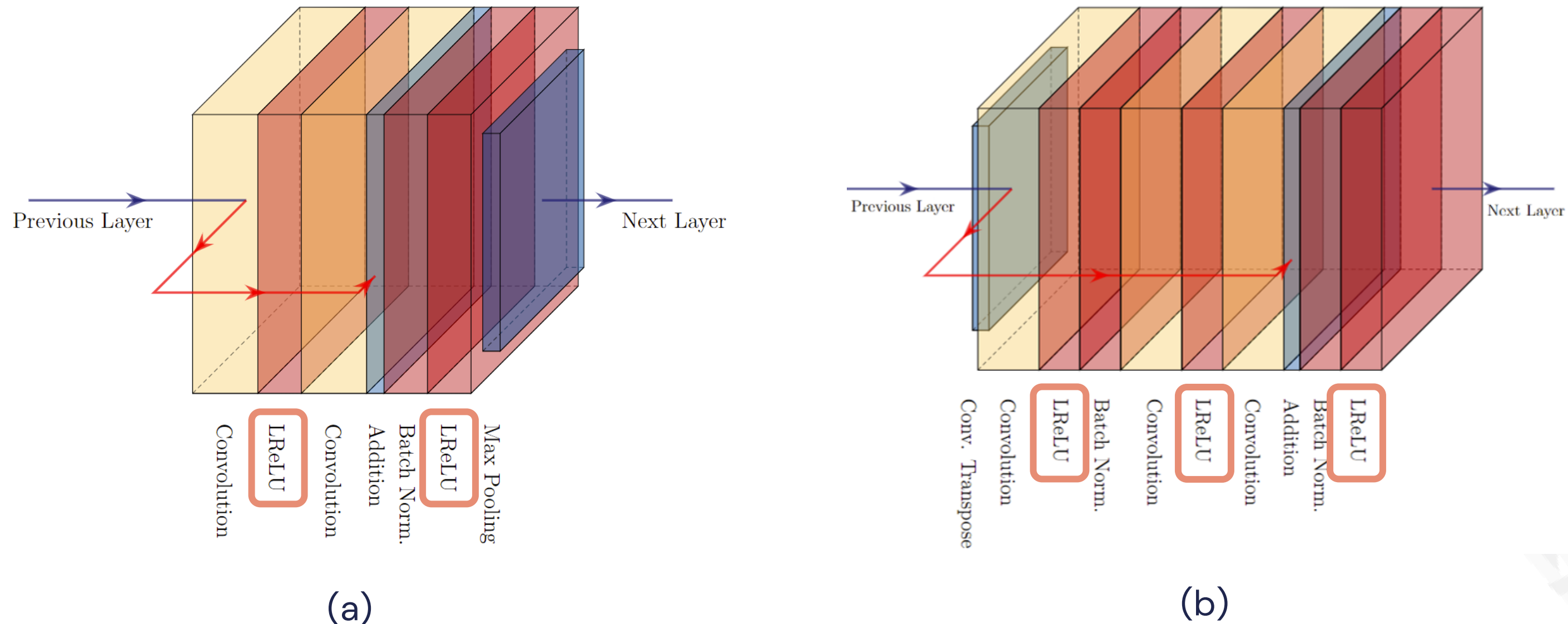


Figure 4.3. Encoder (a) and decoder (b) block in with Leaky ReLU implemented.

Batch normalization (BN) layers are early in the network to normalize the activations. However, it may cause **over-normalization**, which could degrade the shuttlecock detection performance. By applying BN after the addition operation, the input to the activation function is **normalized**.

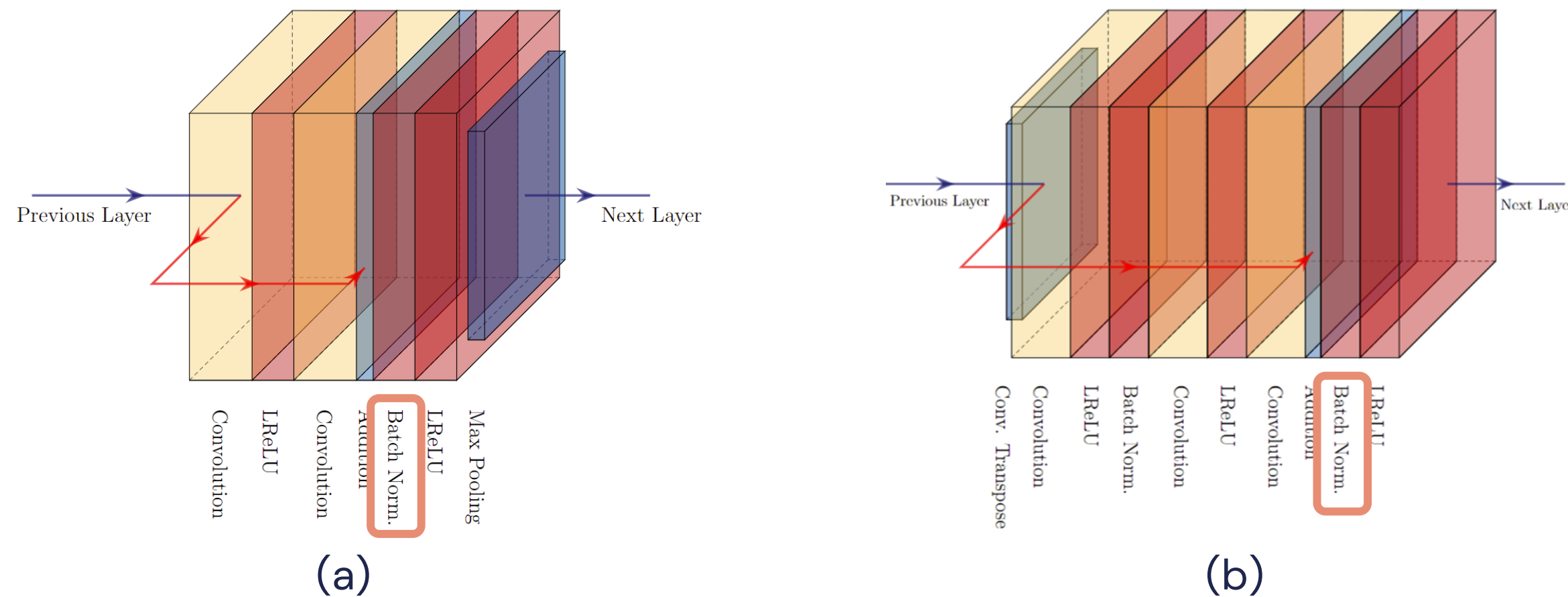


Figure 4.4. Encoder (a) and decoder (b) block with batch normalization placed after residual block.

Chapter 4

# Shuttlecock Detection Dataset



Figure 4.5. Example frames in the shuttlecock detection dataset.



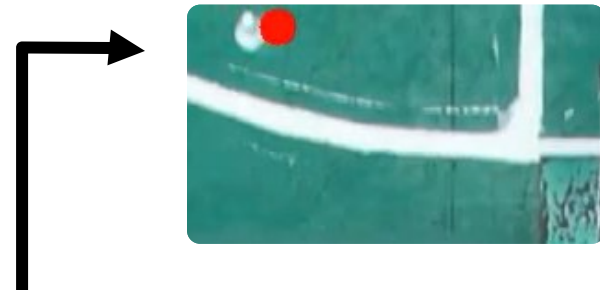
Figure 4.6. Example frames in the shuttlecock detection for testing data.

# Chapter 4

## Shuttlecock Detection

### Result

Baseline



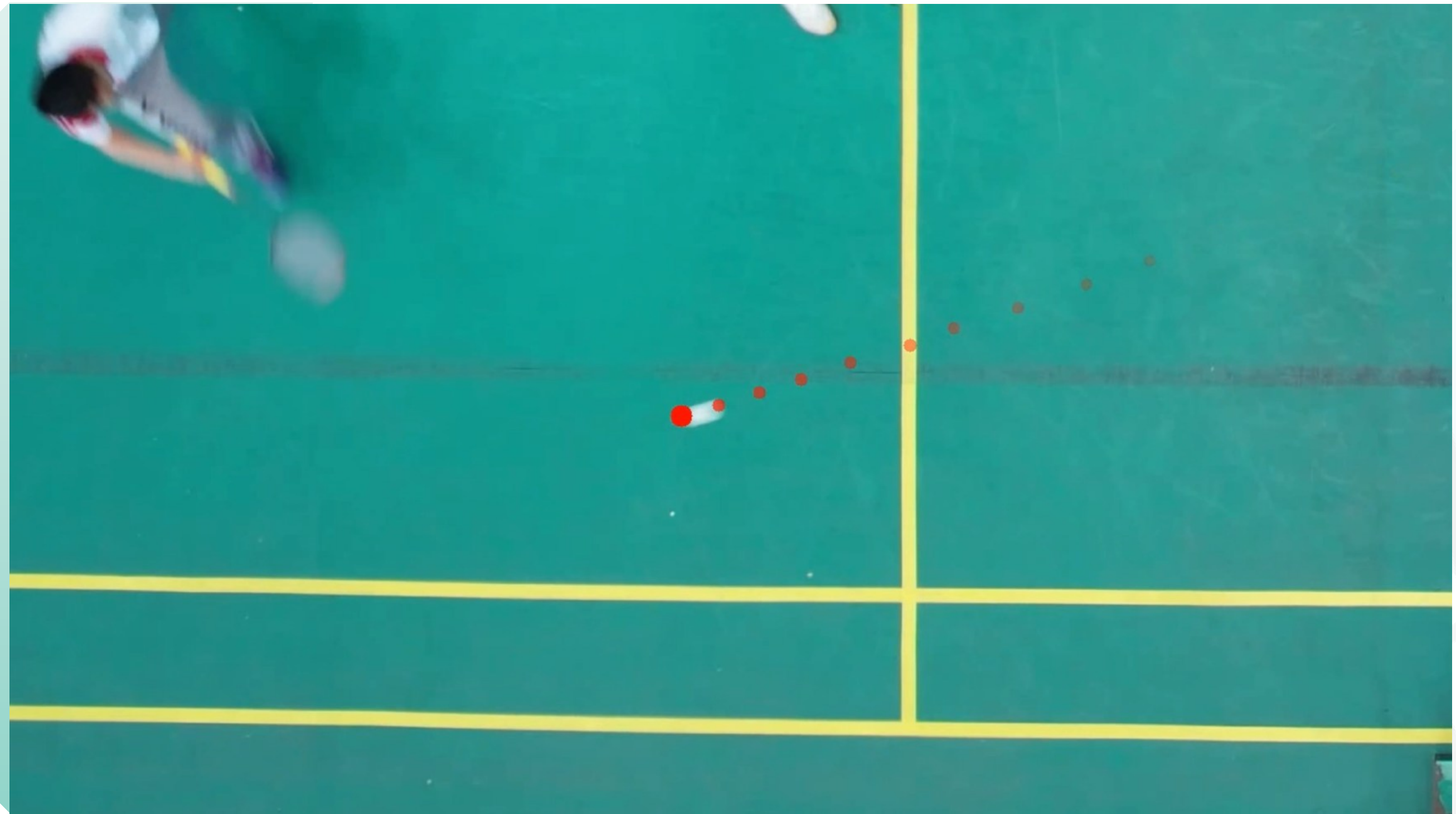
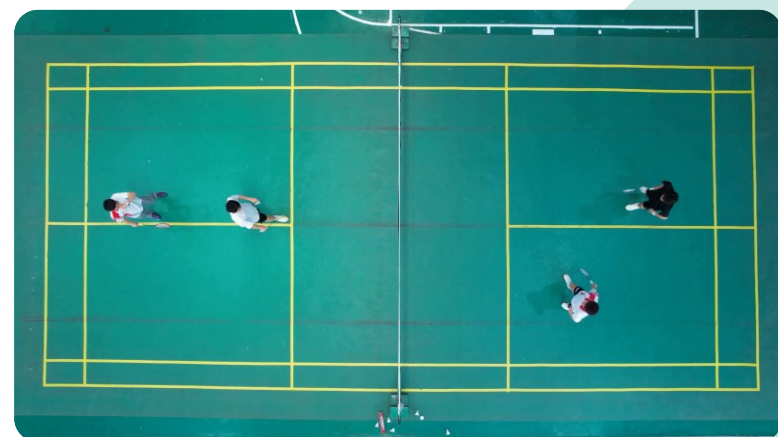
Proposed Method



## Chapter 4

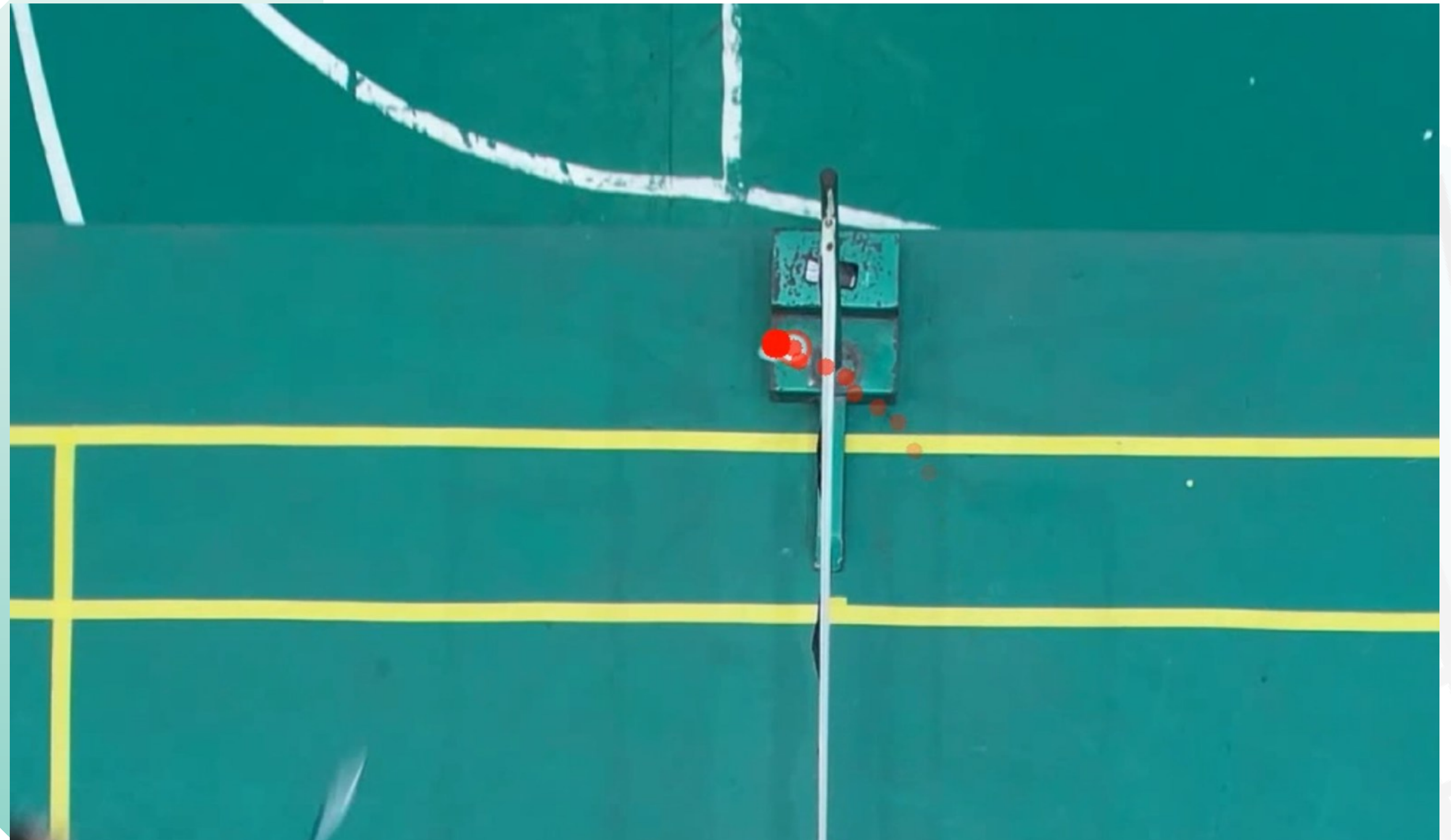
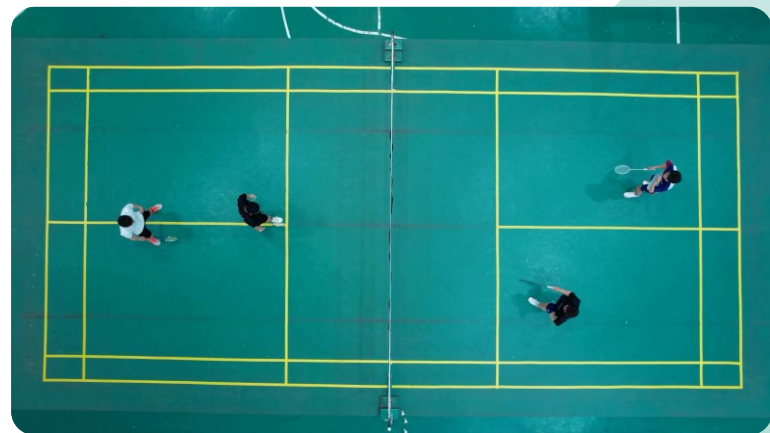
# Shuttlecock Detection

### Result (Cont.)



# Shuttlecock Detection

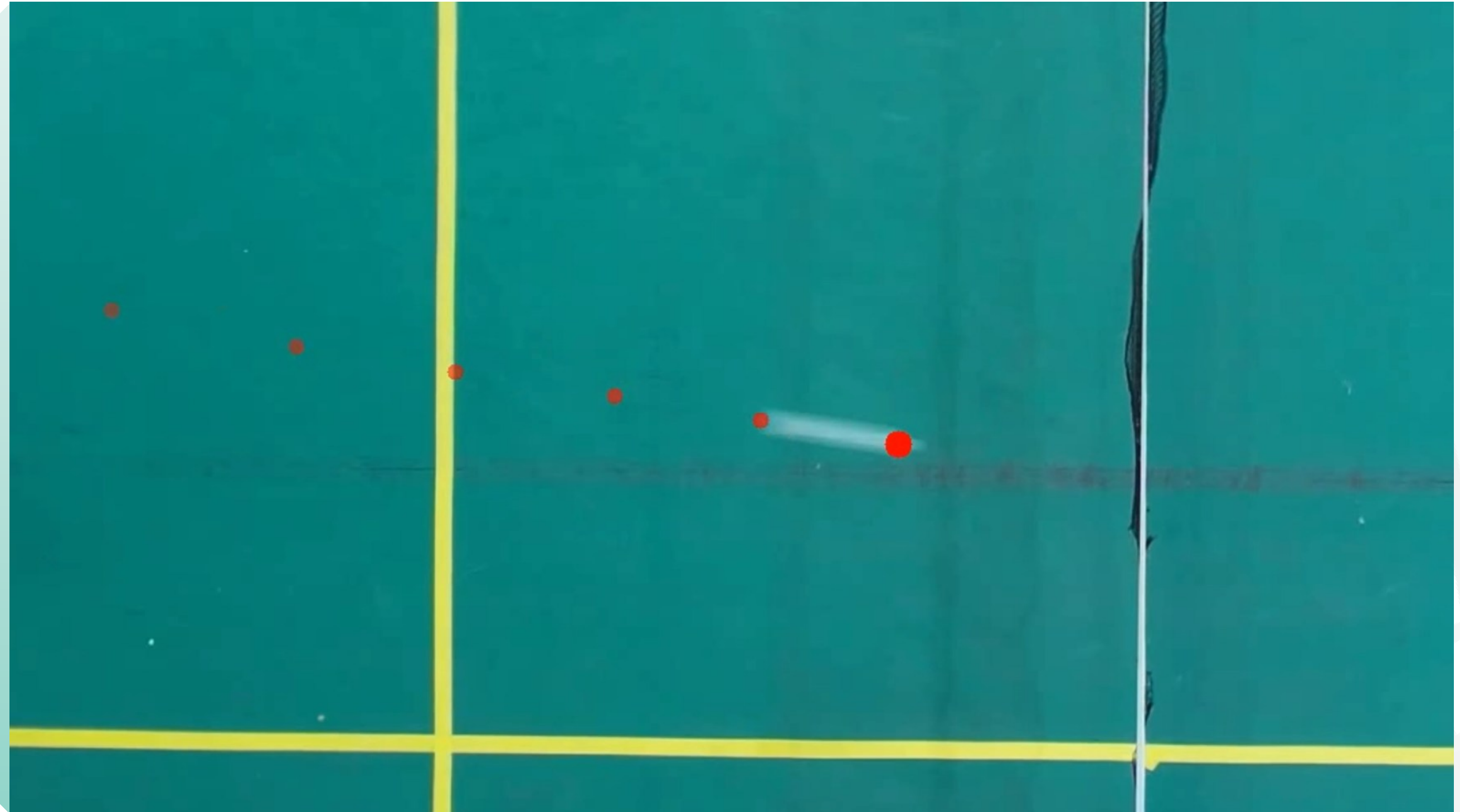
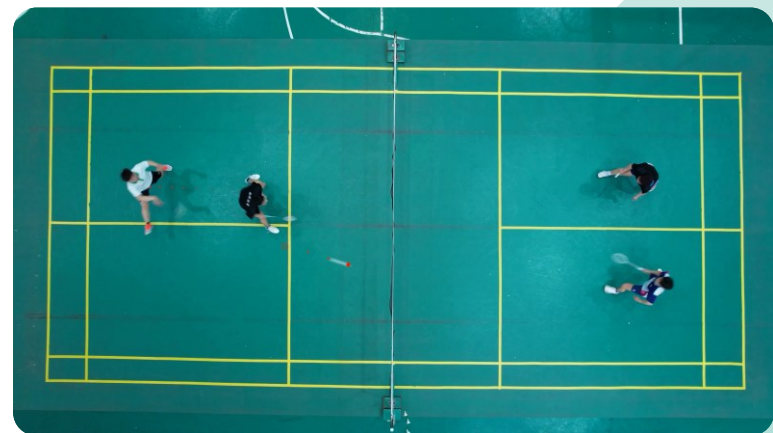
Result : Slow moving shuttlecock (Net Play)



## Chapter 4

# Shuttlecock Detection

Result : Fast moving shuttlecock (Smash)



Chapter 4

# Shuttlecock Detection

Result : Training Loss

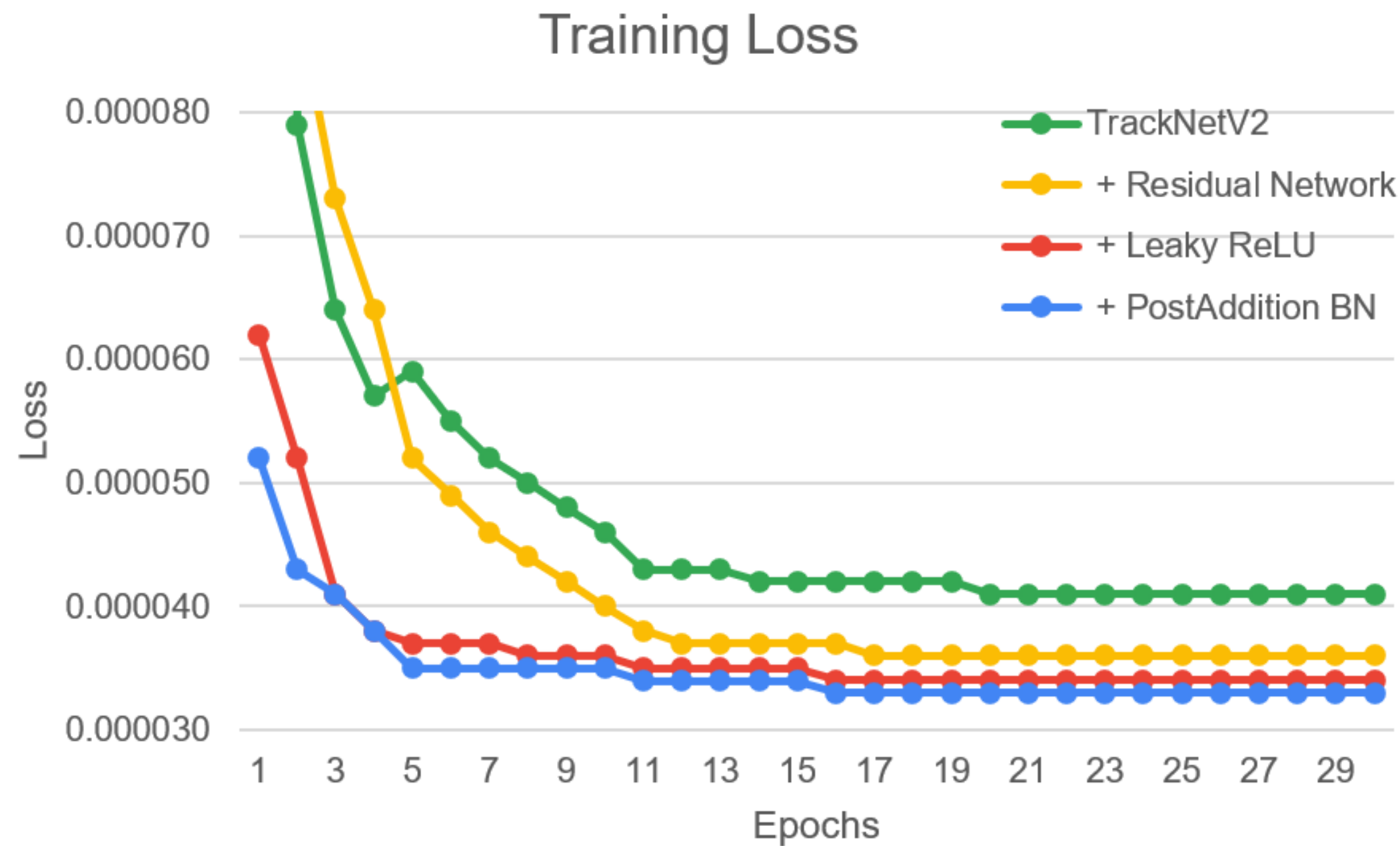


Figure 4.6. Training loss comparison between baseline and proposed method.

Chapter 4

Shuttlecock Detection

Result : Metric Comparison

|        |          | Predicted           |                     |
|--------|----------|---------------------|---------------------|
|        |          | Negative            | Positive            |
| Actual | Negative | True Negative (TN)  | False Positive (FP) |
|        | Positive | False Negative (FN) | True Positive (TP)  |

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

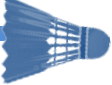
$$F1 = 2 \times \frac{P \times R}{P + R}$$

| Model              | Prec. | Rec.  | F1    | Acc.  |
|--------------------|-------|-------|-------|-------|
| Baseline           | 84.8% | 81.8% | 83.3% | 75.8% |
| + Residual network | 87.4% | 82.0% | 84.6% | 77.6% |
| + Leaky ReLU       | 87.9% | 82.6% | 85.0% | 77.9% |
| + Post-Addition BN | 88.2% | 83.1% | 85.3% | 78.1% |

Table 4.1. Performance comparison between baseline with proposed method.

- Shuttlecock detection accuracy was improved by extending TrackNetV2 with residual learning to stabilize deep feature propagation.
- Leaky ReLU mitigated the dying ReLU problem, while post-residual batch normalization reduced over-normalization effects.
- These architectural refinements consistently increased Precision, Recall, F1-score, and overall accuracy.

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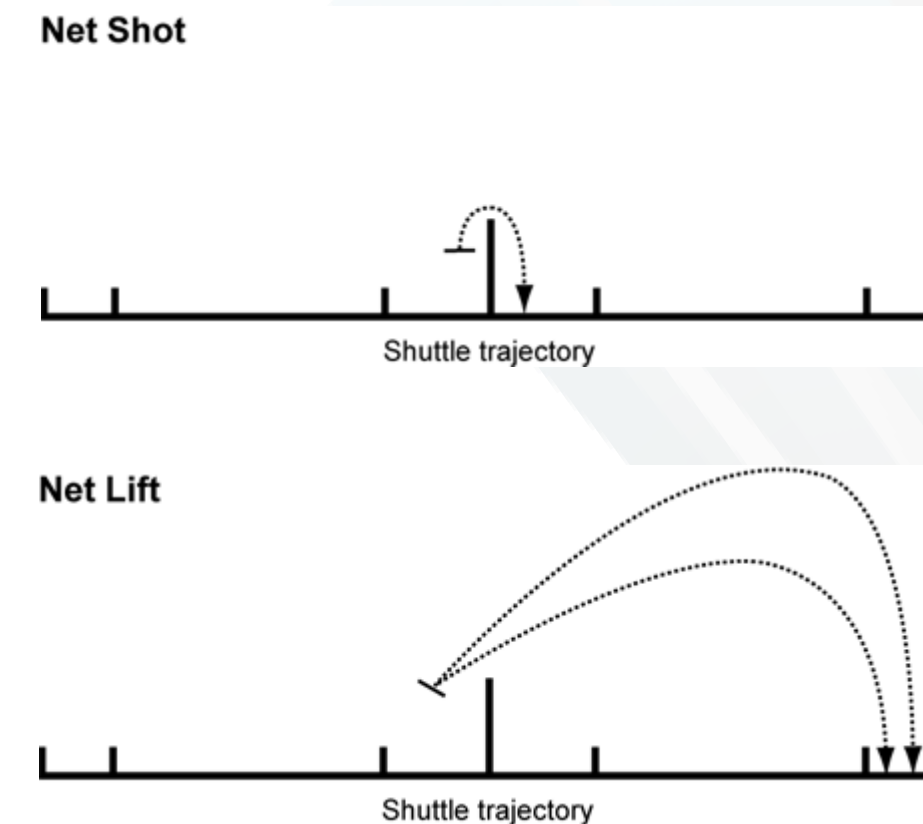
## Chapter 5

# Racket Detection

### Introduction

Why racket matter?

The racket defines the effective interaction point between player and shuttlecock. In net play, **small changes** in racket position strongly **affect** shot outcome. **Racket orientation** reflects intention, timing, and shot readiness.



- Racket detection in badminton videos is challenging due to the **lack of publicly available datasets** that provide explicit racket orientation information. Existing badminton datasets mainly focus on players or shuttlecocks, making racket direction analysis difficult to perform reliably.
- To address this problem, this study constructs a dedicated badminton racket dataset with **oriented bounding box annotations**, including **racket angles**.
- Racket detection is conducted using **YOLOv11-OBB**, and **SORT** is applied to maintain temporal consistency across frames.

ORDB (Oriented Racket Database) is composed of a total of **8,704** images taken from **double** badminton video matches, carefully curated and annotated to provide a diverse range of badminton gameplay scenarios. The dataset is annotated with **rotated bounding boxes**.



Chapter 5

# Racket Detection

Dataset : Annotation

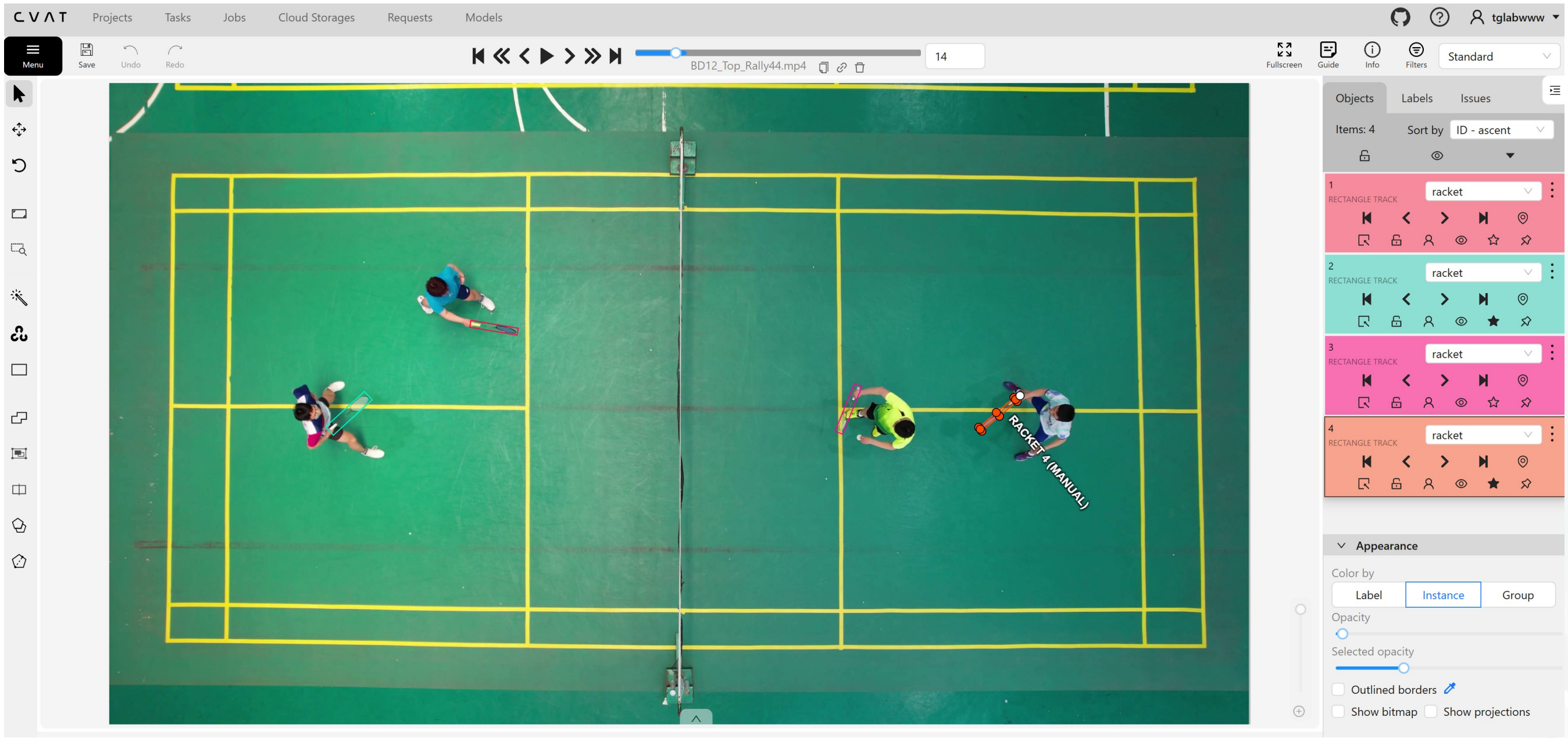


Figure 5.1. Annotation process of oriented racket dataset using CVAT

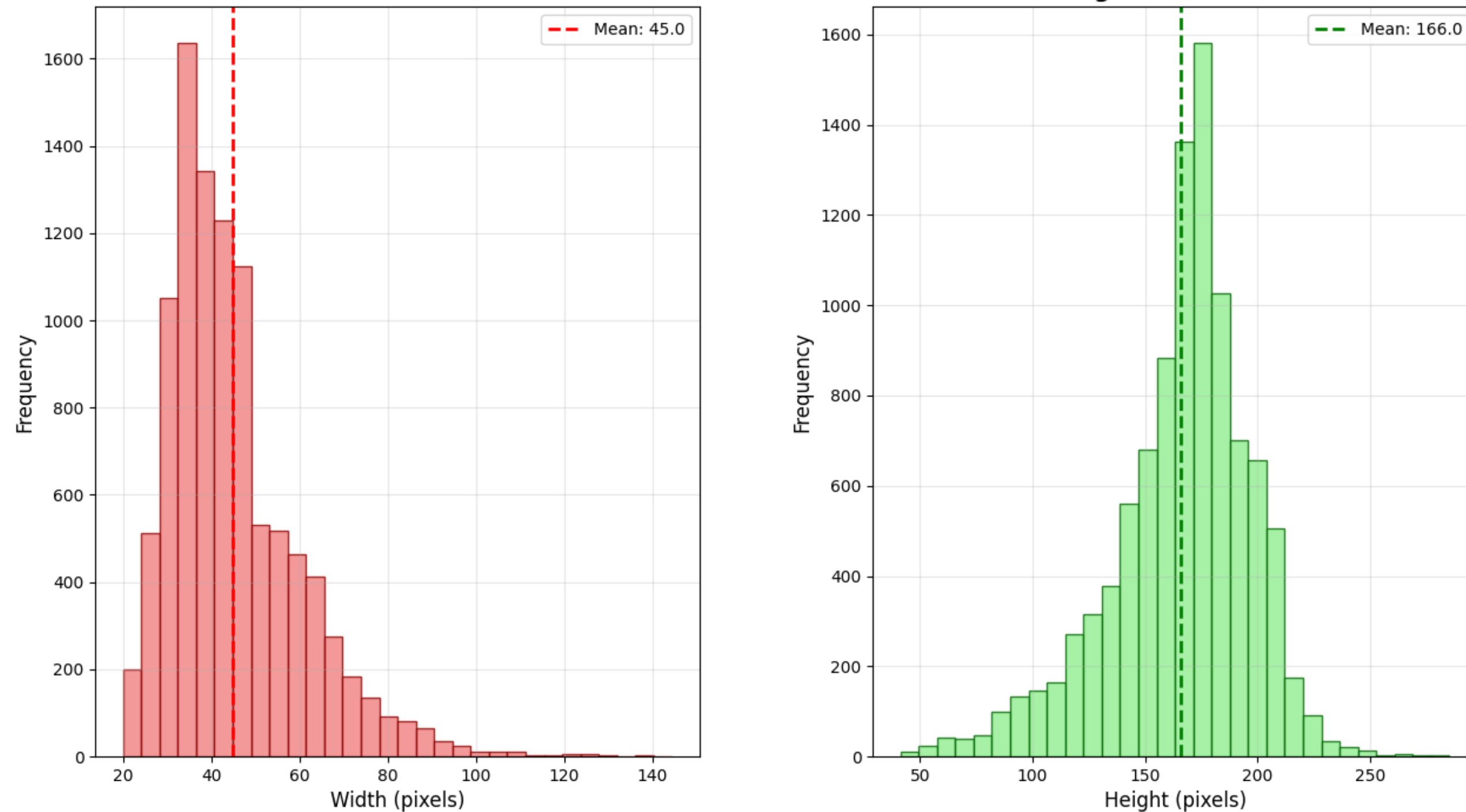


Figure 5.2. Histogram of bounding box width and height

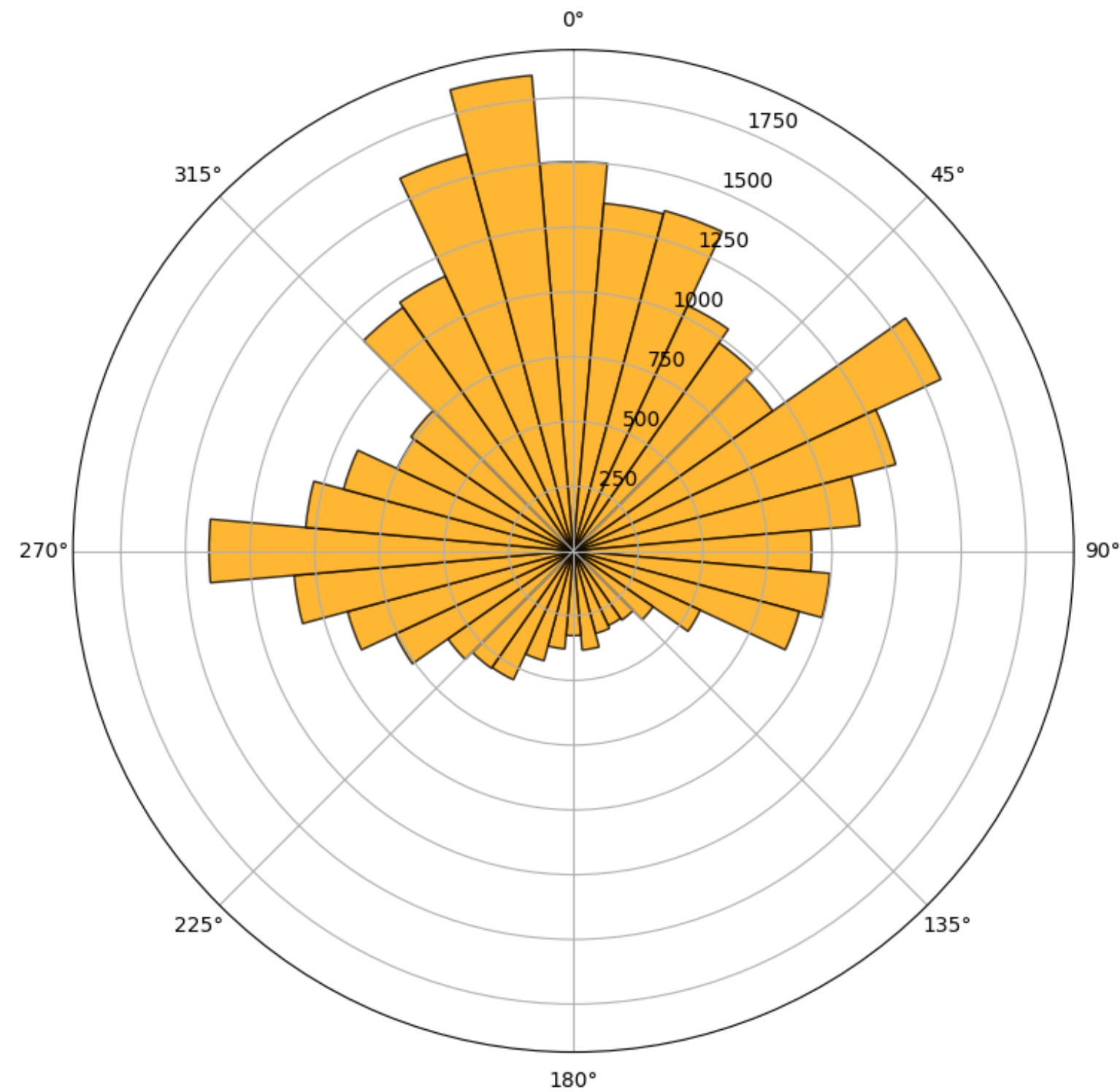


Figure 5.3. Distribution of racket orientation angles in the ORDB dataset.

## Chapter 5

# Racket Detection

### Method

YOLOv11 is used for training and inference, specially for OBB (Oriented Bounding Box) version.

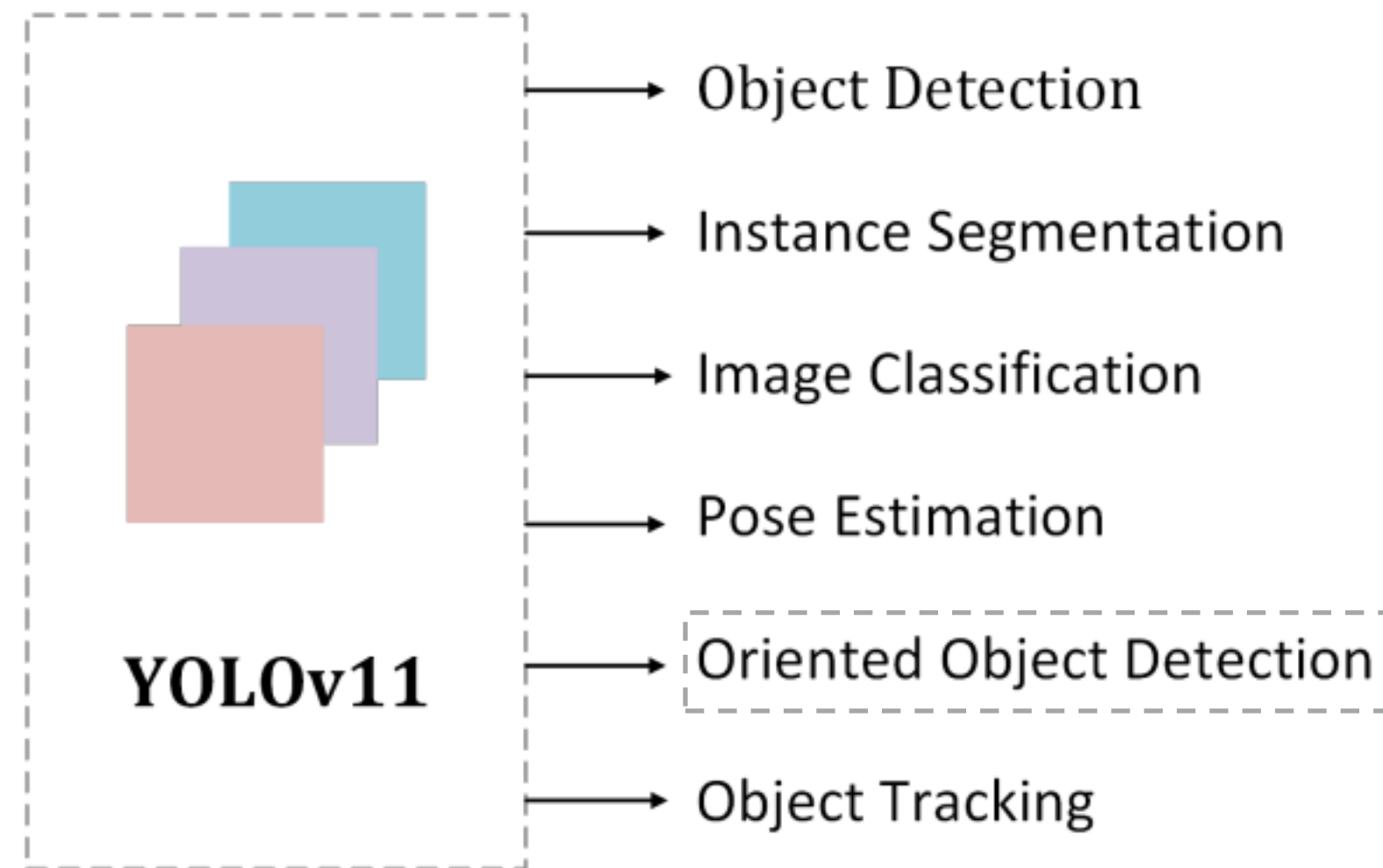


Figure 5.4. YOLOv11, the latest YOLO version as of January 2026, capable of performing oriented object detection.

Qualitative results of YOLOv11-OBB on validation frames. The model successfully detects all four rackets and accurately predicts their positions and orientations.

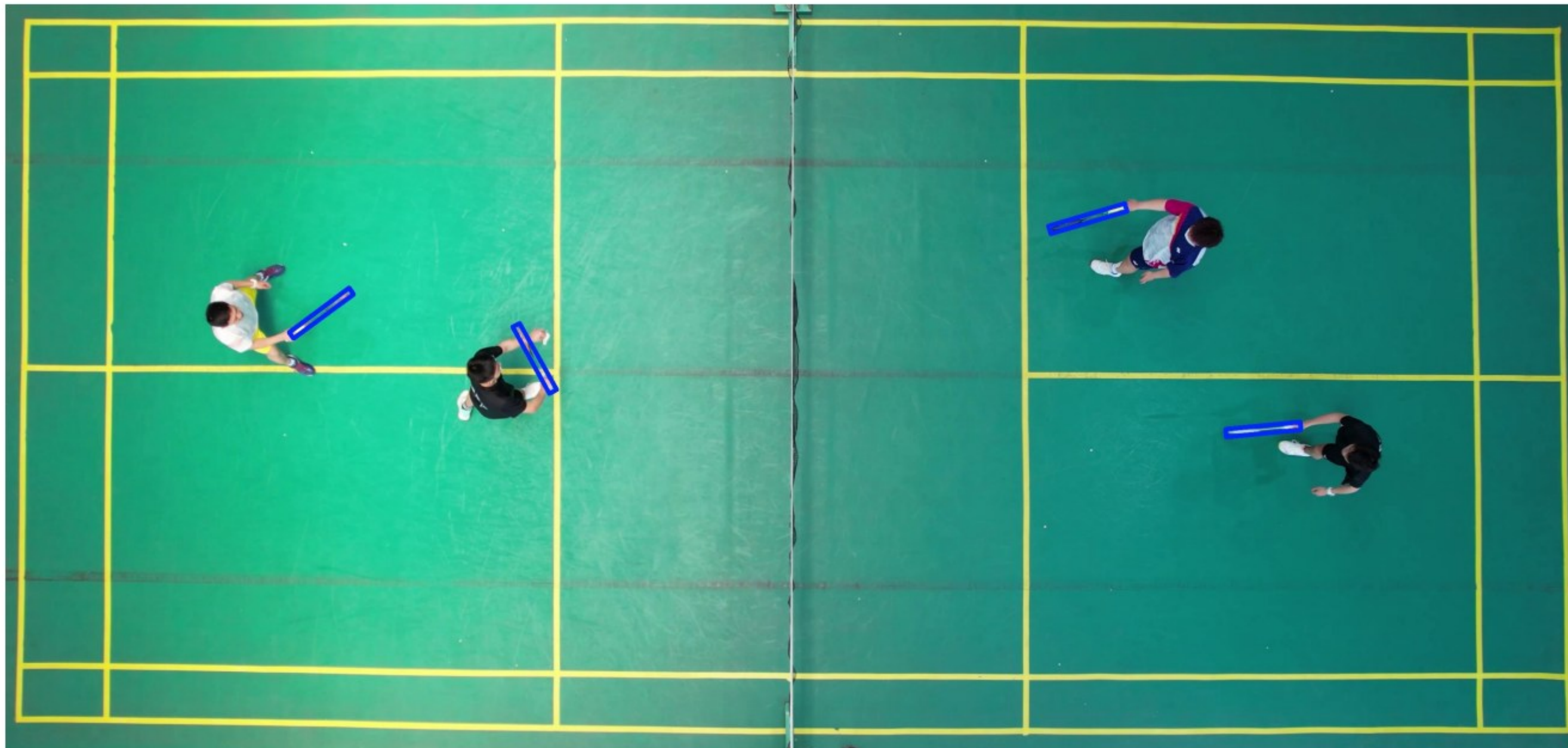
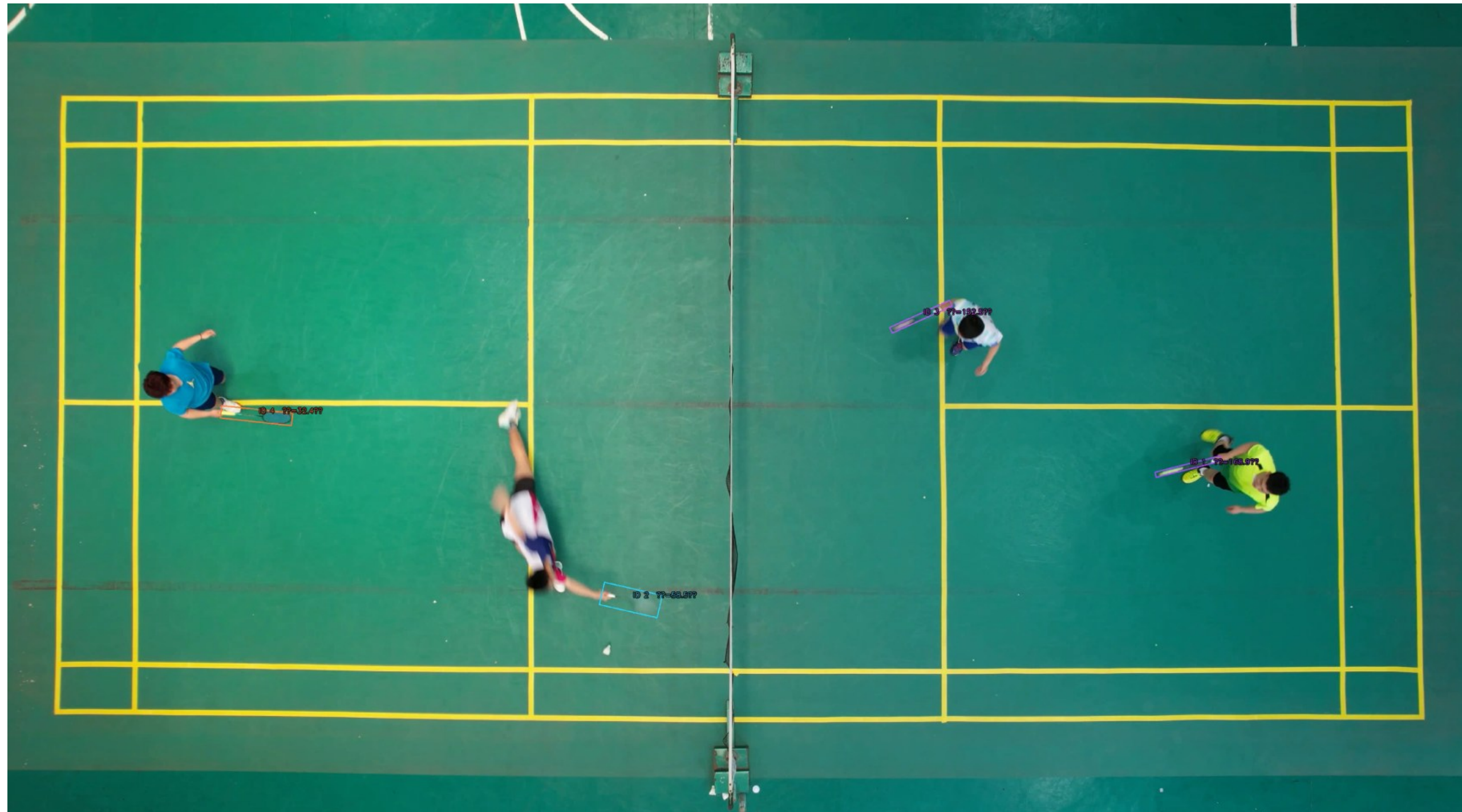
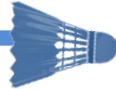


Figure 5.5. The racket was detected successfully with the angle information for every player.

- Racket detection was advanced through the introduction of novel datasets with explicit orientation annotations.
- ORDB employs oriented bounding boxes to model racket pose, enabling rotation-aware detection using YOLOv11-OBB.
- Stable racket trajectories were obtained by integrating SORT tracking, supporting fine-grained stroke analysis.



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## Chapter 6

# Dominant Area

### Introduction

- In badminton doubles, performance depends on how **effectively** pairs control and **share court space** rather than on individual skills alone.
- Rapid role switching during rallies requires **coordinated positioning** to cover open areas and respond to the shuttlecock. Poor spatial coordination often creates gaps that can be exploited by opponents.



- This study addresses this problem by introducing a **dominant area** visualization that estimates court regions reachable by each player based on movement dynamics and shuttlecock context, while also incorporating racket angle information.
- Dominant area represents **which player can reach** a region of the court faster. It captures spatial control that is not visible from positions alone. This information is fundamental for understanding tactics in fast rallies.



Chapter 6

# Dominant Area Method

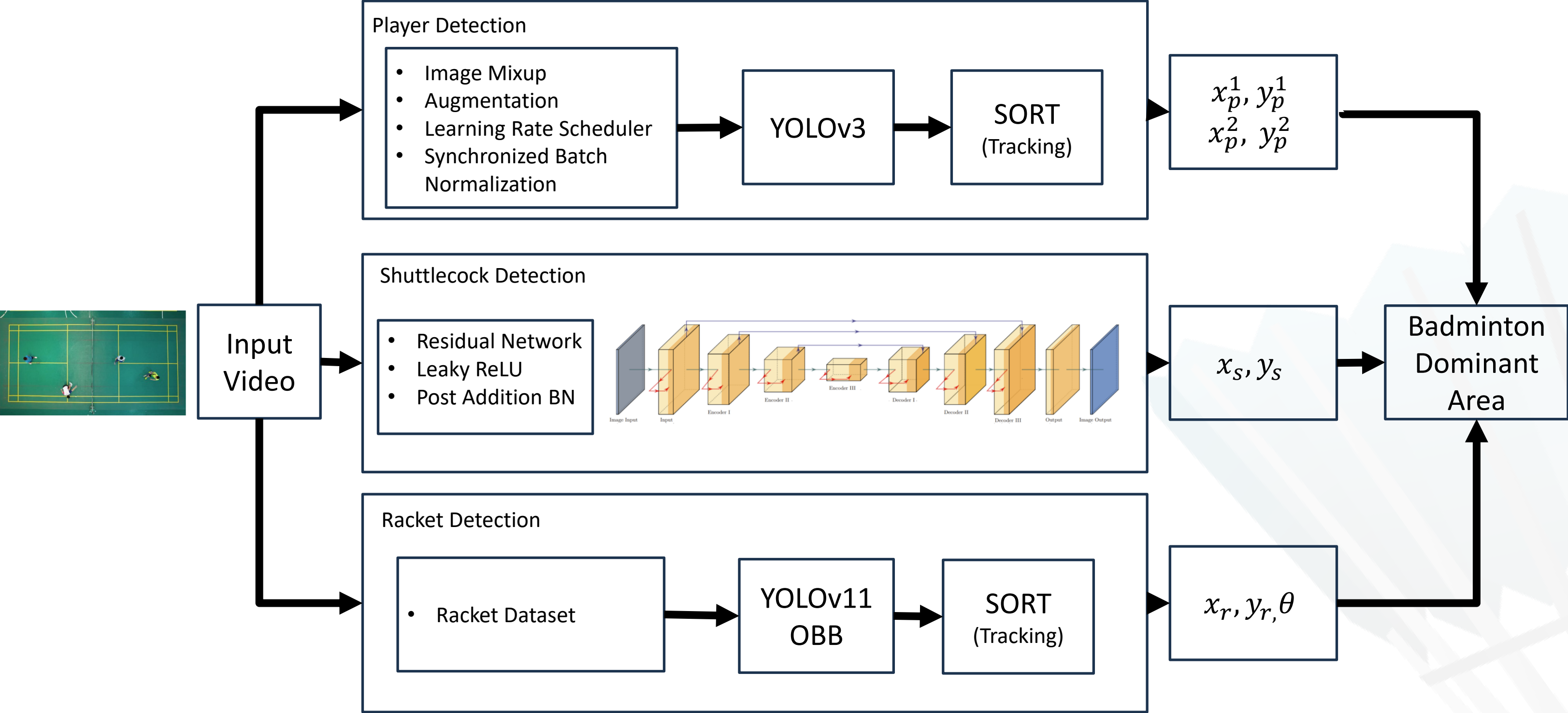
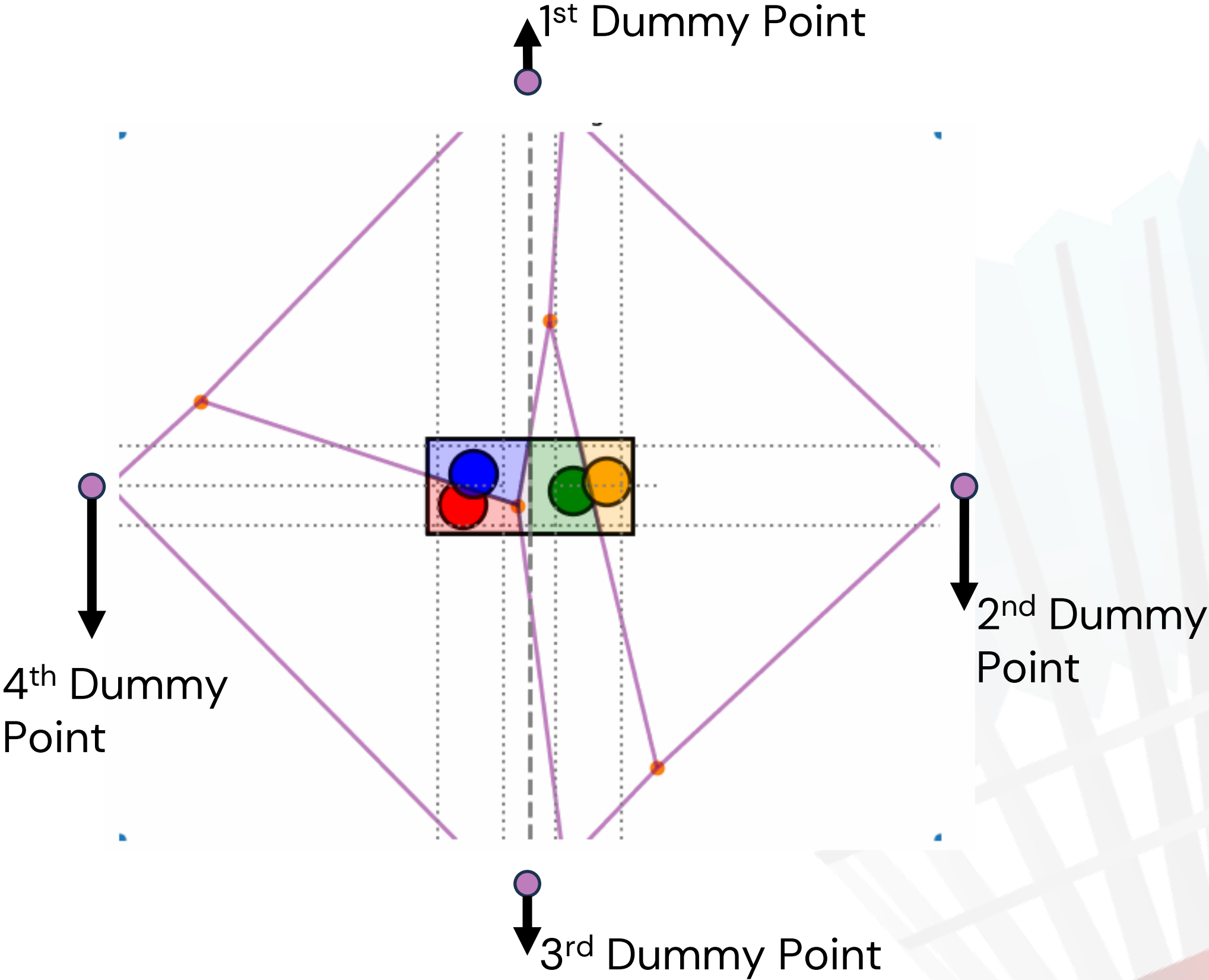
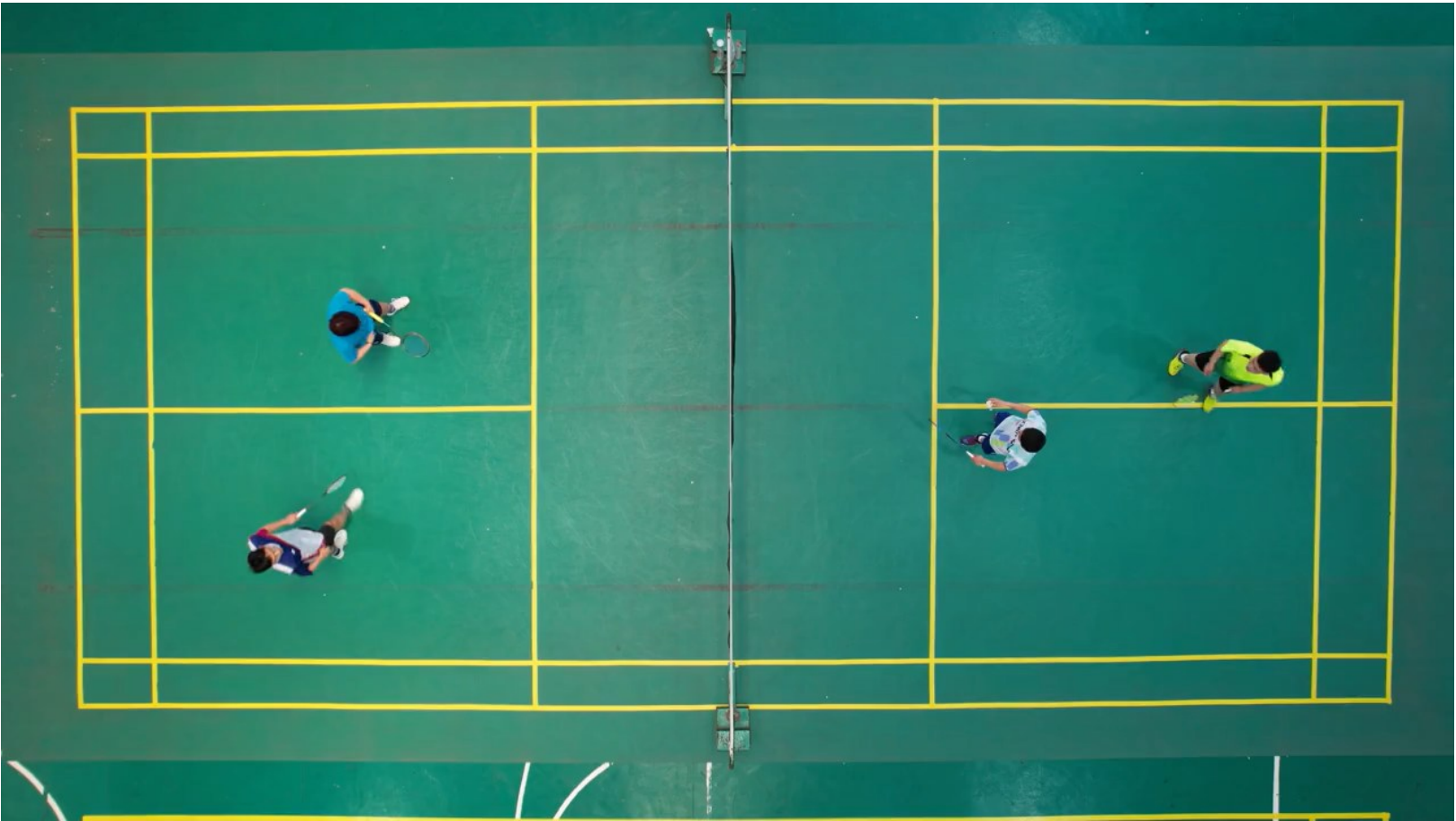


Figure 6.1. Overall pipeline for dominant area estimation in badminton sport analysis

Chapter 6

# Dominant Area

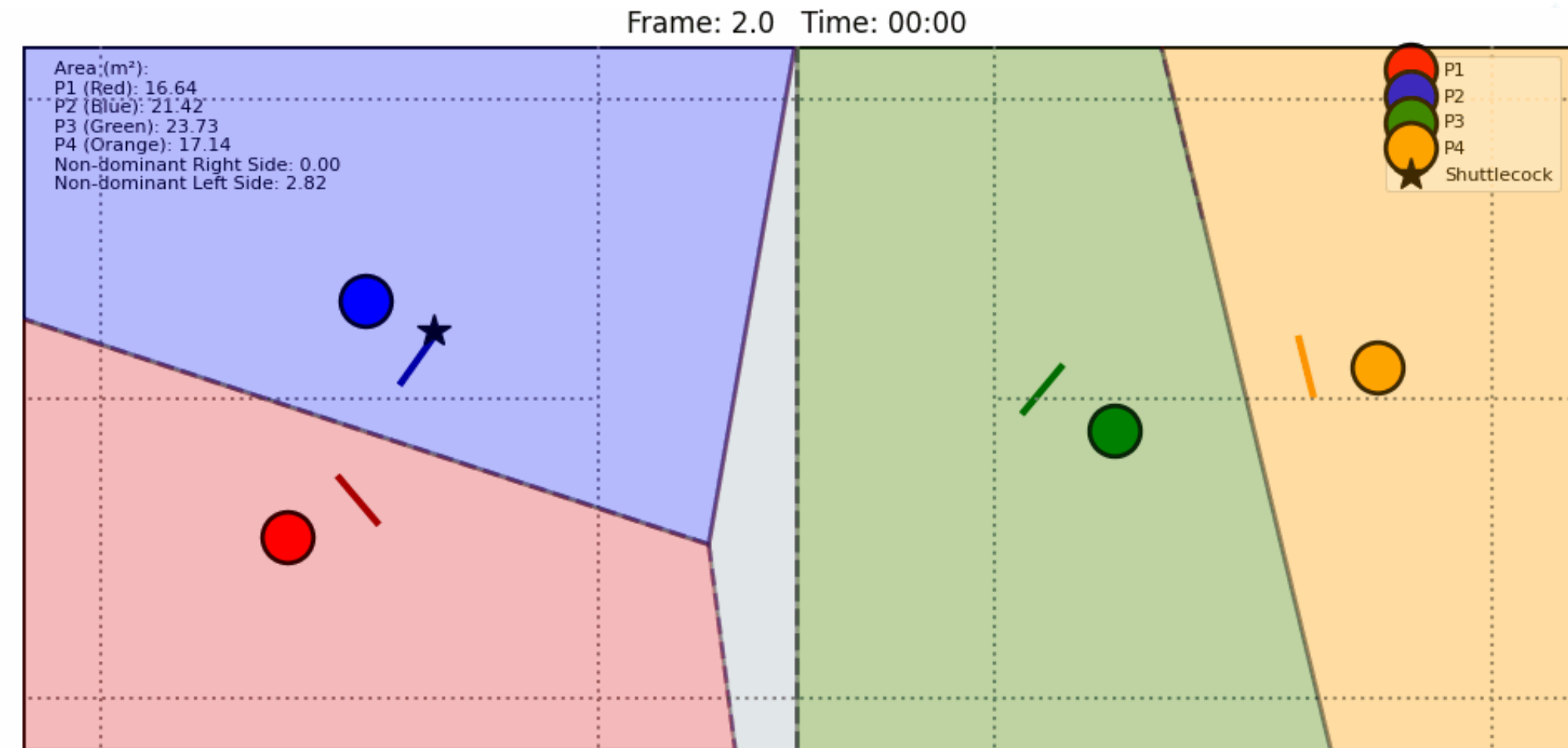
Method : Voronoi Diagram



## Chapter 6

# Dominant Area

### Result



- In doubles, **left-right** formation supports defensive coverage.
- **Front-back** formation is used for attacking and pressure control.
- These roles emerge naturally to reduce reaction time and overlap.

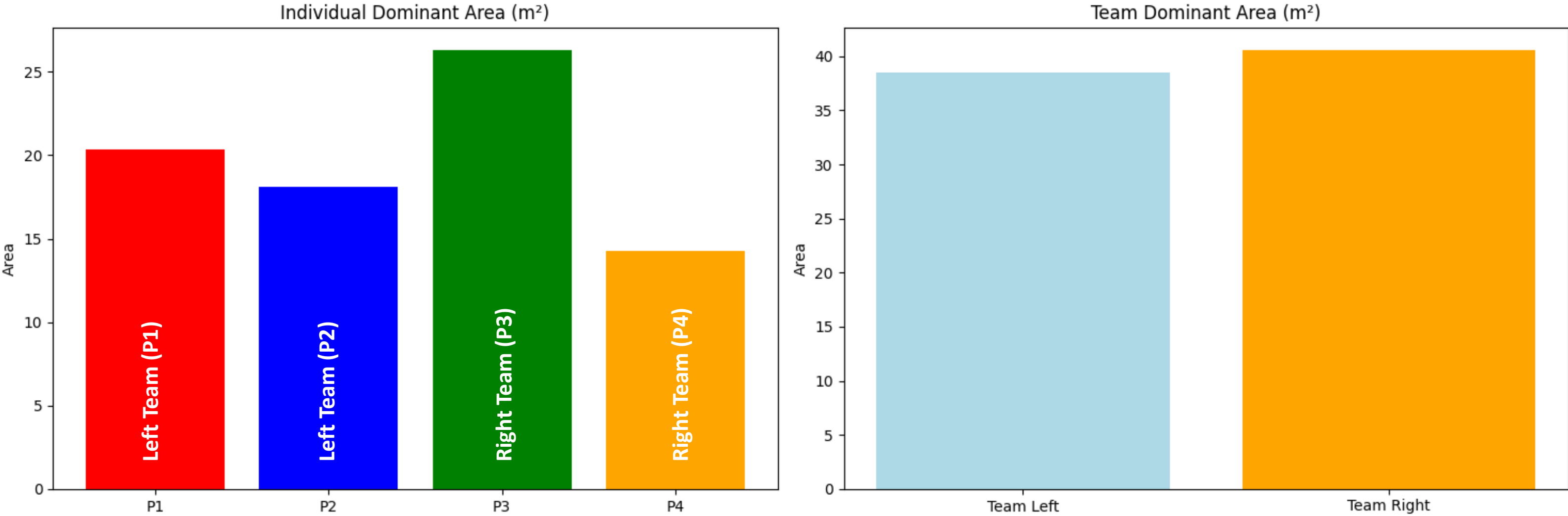


Figure 6.2. Individual and team dominant area in badminton double matches

- The proposed contributions deliver a dominant area estimation by integrating player, shuttlecock, and racket information.
- The visualization reveals tactically relevant regions of area in doubles matches beyond player position alone.
- This approach aligns with sports science concepts of anticipation and effective court coverage.

# Outlines

- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Player Detection
- Chapter 4: Shuttlecock Detection
- Chapter 5: Racket Detection
- Chapter 6: Dominant Area
- Chapter 7: Conclusions and Future Works



- A **short user survey** was conducted to evaluate the clarity and usefulness of badminton video visualizations.
- Participants rated short rally videos based on player positions, shuttlecock movement, racket direction, and dominant area representation using a 1–10 scale.
- The survey captures subjective feedback from badminton players to complement quantitative analysis.



- Participants (57 Responses):
  - Badminton club in Japan (Tachikawa)
  - Badminton club in Thailand (Bangkok)
  - Badminton club in Indonesia (Yogyakarta)

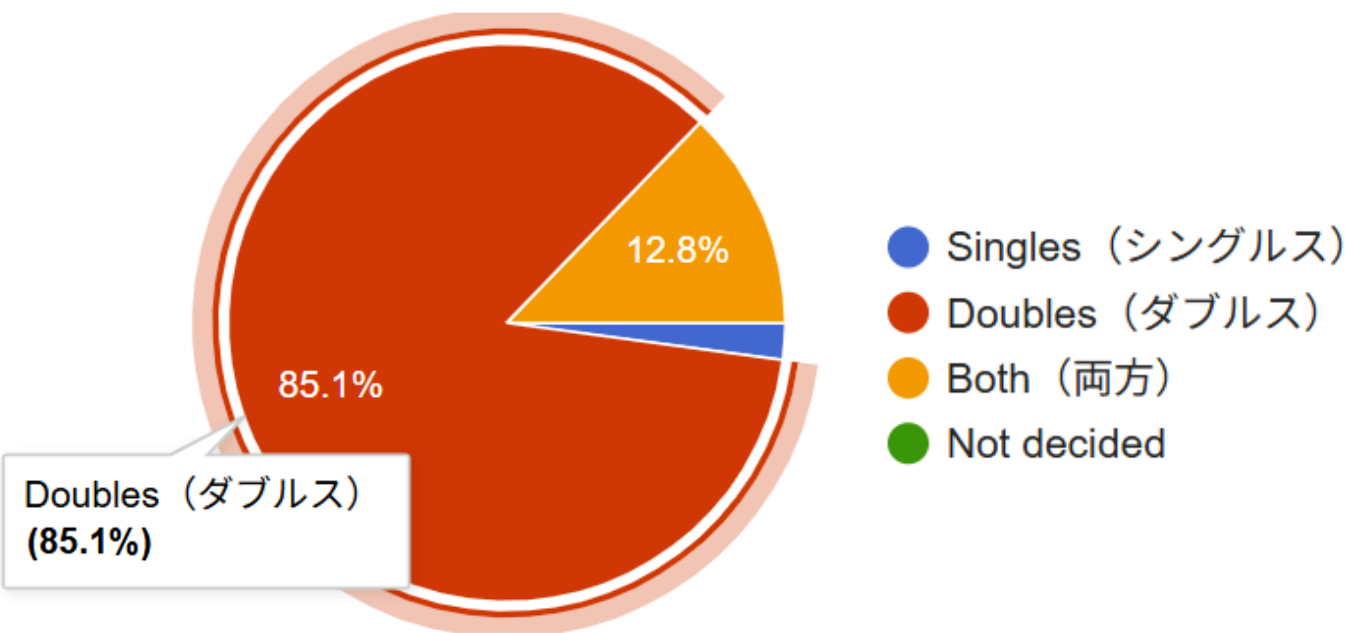


Figure 7.1. Distribution of respondents by duration of badminton playing experience.

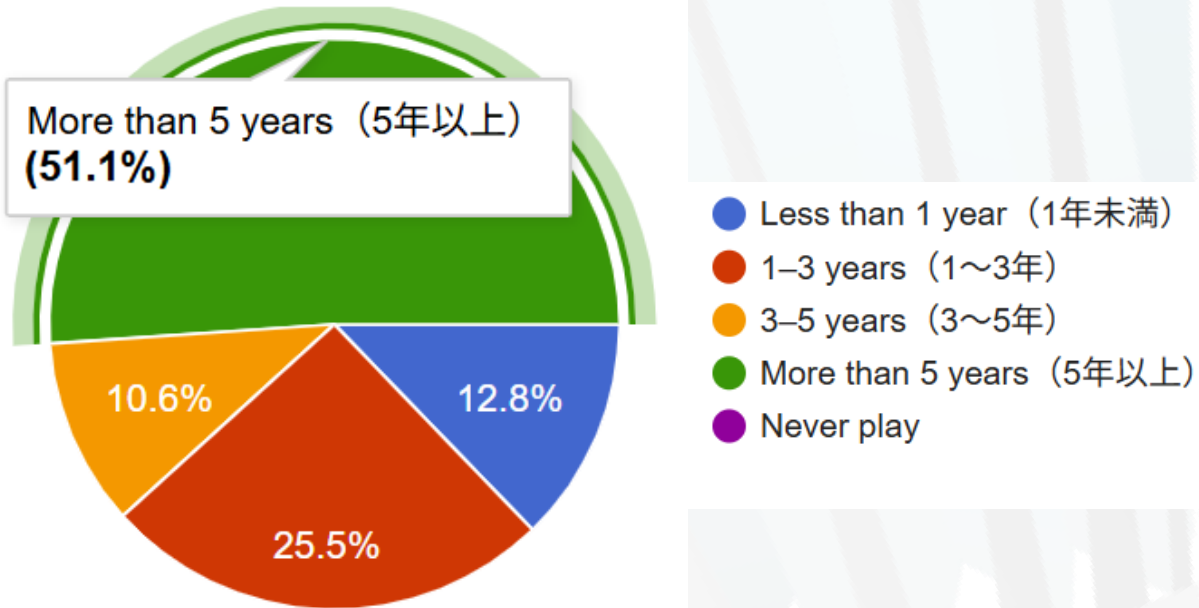


Figure 7.2. Distribution of respondents by usual match type played.

## Conclusions and Future Works

### Survey : User Evaluation Results

User evaluation scores were collected through a survey using a **1–10 rating scale**. Player and Shuttlecock Detection receive consistently high scores, while Racket Detection shows greater variability and the Dominant Area is generally well evaluated.

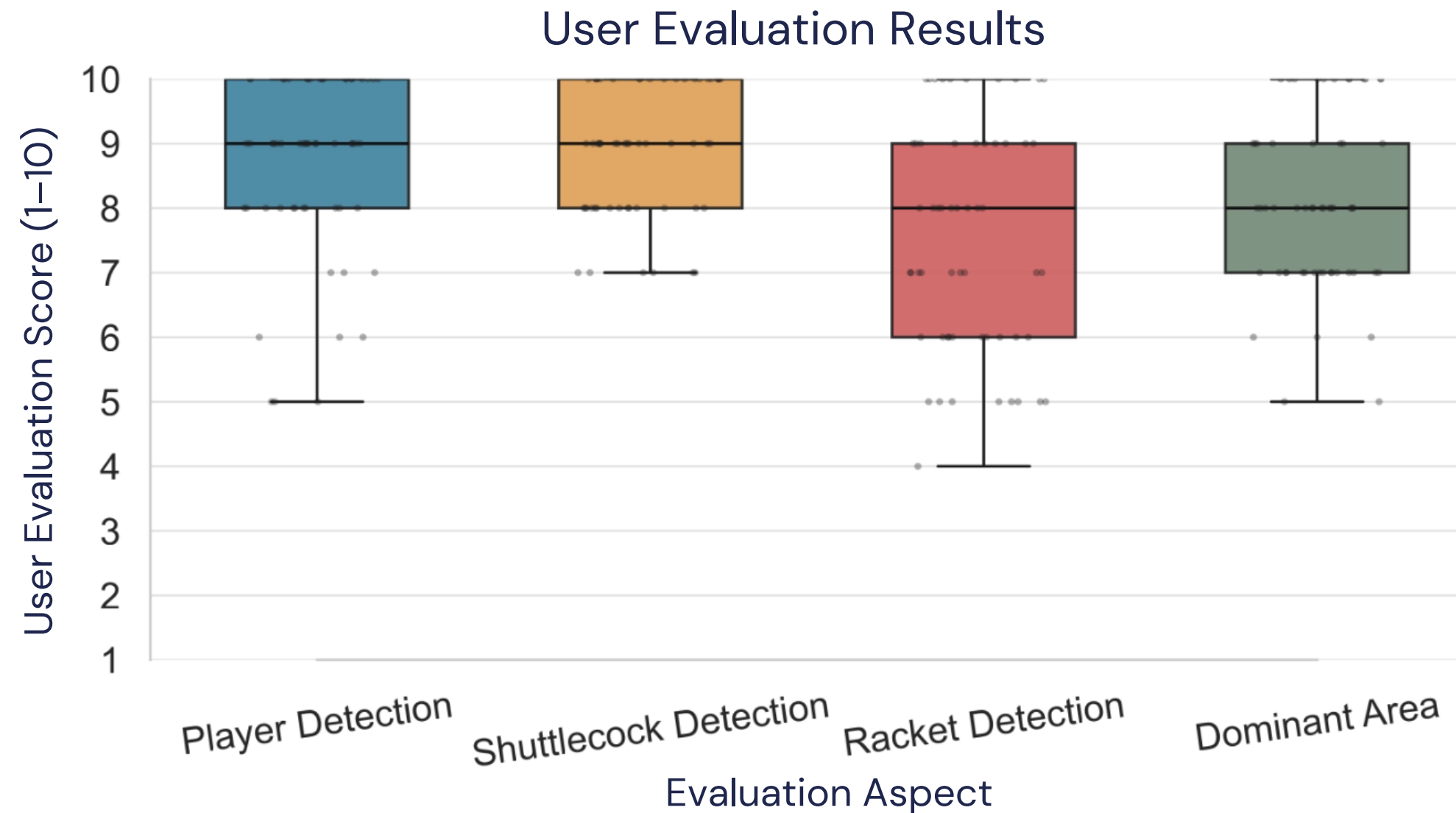


Figure 7.3. Distribution of user evaluation scores (1–10) across evaluation aspects.

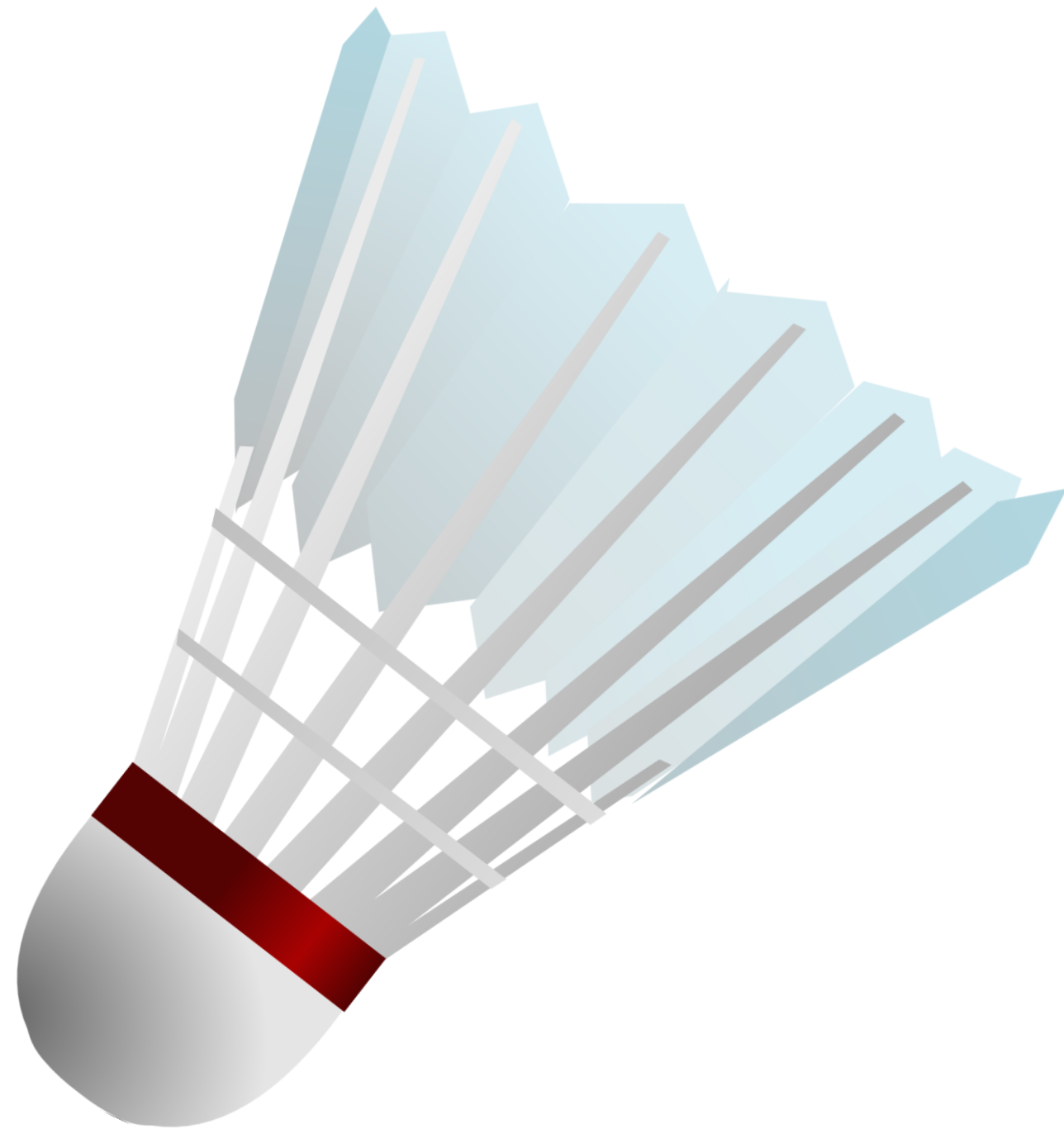
- This dissertation establishes a complete pipeline for badminton sports analysis, including **player detection, shuttlecock trajectory estimation, racket detection, and dominant area estimation**.
- Player detection was improved through **heuristic YOLOv3 training**, and tracking was implemented using **SORT**.
- Shuttlecock detection was enhanced by extending TrackNetV2 with **residual learning** to mitigate vanishing gradients, **Leaky ReLU** to avoid the dying ReLU problem, and **post-residual batch normalization** to prevent over-normalization.

- Racket detection was advanced by creating novel datasets: **ORDB** with **oriented bounding boxes** to explicitly model racket orientation in the detection results, detection was performed using **YOLOv11-OBB** and tracking was conducted with **SORT**.
- Together, these contributions deliver a **Dominant Area** estimation integrated players, shuttlecock, and racket data, providing new insights into tactical regions in double matches. It aligns with sports science concepts of anticipation and court coverage.
- The user evaluation results show median scores of **9** for Player Detection and Shuttlecock Detection, and **8** for Racket Detection and the Dominant Area, indicating generally positive user perception across all aspects.

- Future work focuses on improving dominant area estimation by considering **temporal factors** such as **reaction delay**, and interactions between players to the **shuttlecock natural movement**, so the dominant area during fast rallies can be represented more realistically.
- The framework can be extended to analyze **tactical behavior** in doubles play, particularly by examining player roles in the **front and back court** and the **coordination between teammates**.
- In sport science perspective, future work can integrate **fatigue**, and **player coordination dynamics**.

# Publications

| No | Event                                                                                                     | Date              | Type                     | Title                                                                                           |
|----|-----------------------------------------------------------------------------------------------------------|-------------------|--------------------------|-------------------------------------------------------------------------------------------------|
| 1  | Institute of Image Information and Television Engineers (ITE) Conference 2025 – Yokohama, Japan           | December 18, 2025 | Local Conference         | Integrating Multi-Object Information for Dominant Area Analysis in Badminton                    |
| 2  | ACM MMSports 2025 – ACM International Workshop on Multimedia Content Analysis in Sports - Dublin, Ireland | October 27, 2025  | International Conference | Oriented Racket Database (ORDB): An Annotated Dataset for Rotated Racket Detection in Badminton |
| 3  | VISAPP 2025 – International Conference on Computer Vision Theory and Applications – Porto, Portugal       | February 27, 2025 | International Conference | RacketDB: A Comprehensive Dataset for Detection of Badminton Rackets                            |
| 4  | Institute of Image Information and Television Engineers (ITE) Conference 2024 – Tsukuba, Japan            | March 21, 2024    | Local Conference         | Bunch of Tricks for Improving Shuttlecock Detection from Badminton Videos                       |
| 5  | IEECON 2024 International Electrical Engineering Congress - Bangkok, Thailand                             | March 7, 2024     | International Conference | Bag of Tricks Toward Enhanced Shuttlecock Detection from Badminton Videos                       |
| 6  | International Journal on Informatics Visualization (JOIV)                                                 | December 15, 2023 | Journal                  | Shuttlecock Detection Using Residual Learning in U-Net Architecture                             |
| 7  | ICIST 2023 International Conference on Information and Science Technology – Semarang, Indonesia           | December 15, 2023 | International Conference | Shuttlecock Detection Using Residual Learning in U-Net Architecture                             |
| 8  | Institute of Image Information and Television Engineers (ITE) Conference 2023 – Hokkaido, Japan           | February 21, 2023 | Local Conference         | A Residual U-Net Architecture for Shuttlecock Detection                                         |
| 9  | Institute of Image Information and Television Engineers (ITE) Conference 2022 – Tokyo, Japan              | December 23, 2022 | Local Conference         | Badminton Player Positioning Comparison using Heatmap Data During BWF World Championship 2021   |
| 10 | International Journal on Informatics Visualization (JOIV)                                                 | December 14, 2022 | Journal                  | Improving Badminton Player Detection Using YOLOv3 with Different Training Heuristic             |
| 11 | ICIST 2022 International Conference on Information and Science Technology – Semarang, Indonesia           | December 14, 2022 | International Conference | Improving Badminton Player Detection Using YOLOv3 with Different Training Heuristic             |
| 12 | Institute of Image Information and Television Engineers (ITE) Conference 2022 – Fukushima, Japan          | August 24, 2022   | Local Conference         | Badminton Player Detection and Heatmap Visualization Based on Convolutional Neural Networks     |
| 13 | IES 2022 International Electronics Symposium – Surabaya, Indonesia                                        | August 9, 2022    | International Conference | Heatmap Visualization and Badminton Player Detection using Convolutional Neural Network         |



Thank you.

ありがとうございます